

Hydro Series Cooling User Guide

2026/07/13

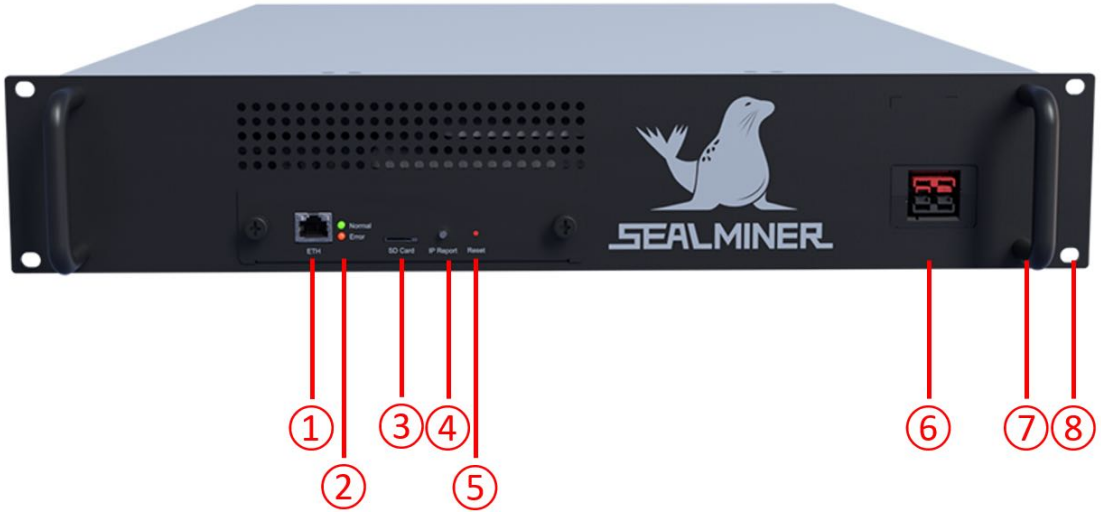
www.bitdeer.com

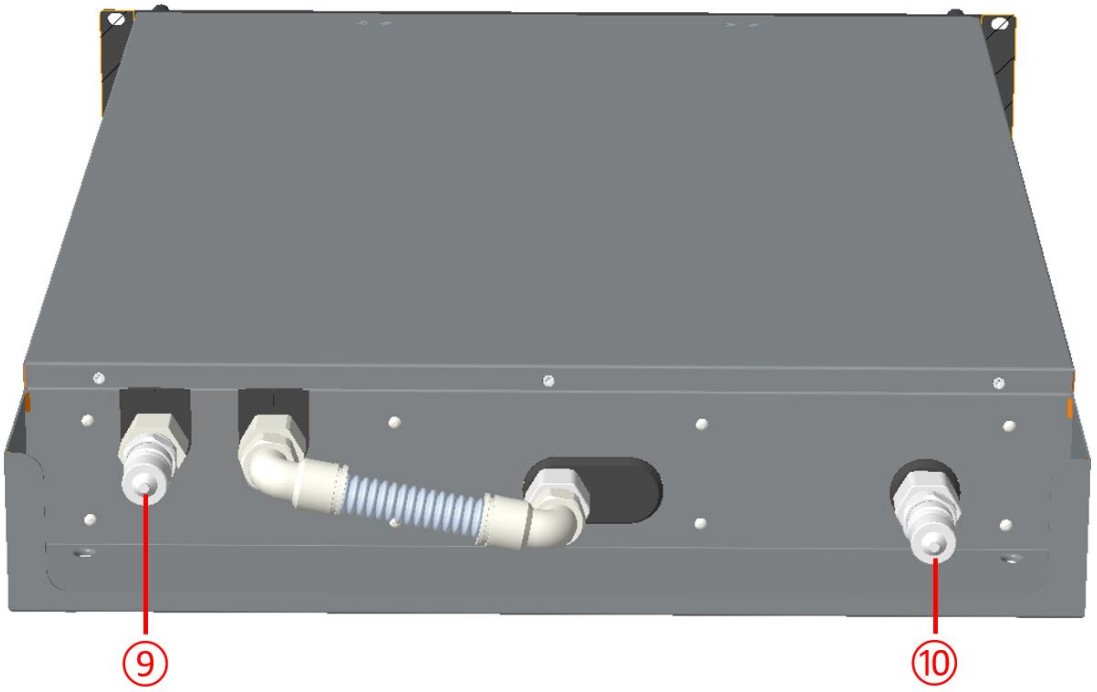
Table of Contents

1. Product Introduction **3**
2. Product And Environmental Parameters **4**
3. Cooling system design requirements **10**
3.1 Server Inlet Temperature Control 10
3.2 Server Inlet Flow Control 10
3.3 Server water inlet pressure 10
3.4 Server coolant medium 10
3.5 Circulation cooling system 11
4. Product installation and operation instructions **13**
4.1 Installation location 13
4.2 Installation Steps 14
4.3 Uninstallation Steps 18
4.4 Note 21
5. Important Notes for Powering On and Off the Product **23**
5.1 Pre-Startup Checklist 23
5.2 Startup Procedure 24
5.3 Shutdown Precautions 24
6. Product Accessory Specifications **25**
7. Network Configuration **28**
7.1 Server Configuration Device List 28
7.2 Server Network Environment 29
8. Data Configuration **29**
8.1 Query the Dynamic IP Address Assigned to the Server 29
8.2 Mining pool & miner data 31
8.3 Configuring a static IP address for the server 32
9. Server Operation Status Check **33**
10. Batch Configuration **34**
11.Disassembly and installation **34**
11.1 Removing the Control Board 34
11.2 Disassembling the Hash Board 35
11.3 Removing the Power Supply 39
12. After-Sales Service **41**
13. After-sales warranty Fee terms **43**
14. Regulatory and Compliance information **45**
14.1 United States Federal Communications Commission (FCC) 45
14.2 Innovation, Science and Economic Development (ISED) Canada 47
14.3 European Union (CE) & WEEE 48
14.4 Safety & Power Specifications 49
14.5 Product Identification & Labelling 50

1. Product Introduction

This operation manual is applicable to the following models: A2 Hyd, A2 Pro Hydro, A3 Hydro, A3 Pro Hydro, A4 Pro Hydro, and A4 Ultra Hydro. (The parameters in this guide are applicable to the general conditions of the listed models. If there are differences in technical parameters for specific models, please refer to the latest data on the official Bitdeer website. If you have any questions or need further assistance, please contact the Bitdeer customer support team.)





①Ethernet port
②Indicator light
③SD card slot

④Find IP button
⑤Reset button
⑥Power port
⑦Handle
⑧Fixing hole
⑨Water inlet (blue mark)
⑩Water outlet (red mark).

2. Product And Environmental Parameters

Table 1: Product Parameters

Specifications		
Power Supply	Power Supply Model	AC380V~480V, 3W+Ground, input 10KW , 17A MAX, 50/60Hz
Physical Parameters	Product Dimensions, W*D*H	482*665*86mm
	Packing Dimensions, W*D*H	567*787*203mm
	Product Colour	Front Panel：Black Exterior：Grey
	Product Weight	Net Weight：20.5kg Weight Including Packaging：23kg
		Net Weight：21kg Weight Including Packaging：23.5kg (A4 Ultra Hydro)
	Coolant Capacity (Single Server)	Approximately 0.7L
Material	Parts In Direct Contact With Coolant	Aluminum Alloy, Stainless Steel, EPDM, FEP

Environmental Parameters	Coolant Temperature	1. Working temperature (Medium: water; specific inlet fluid temperature please refer to the official website for relevant technical parameters): 20℃~55℃@normal mode; 20℃~50℃@high hash mode; 2. Liquid inlet temperature control accuracy ±2℃; 3. Storage and transportation temperature: -30~70℃ (Note: Please drain the liquid in the server during storage and transportation; if not, ensure that the server is above the freezing point of the liquid)
	Coolant Flow Rate	1. Flow rate ≥ 10.2L/min; 2. Accuracy ±10%; 3. 10.2L/min corresponds to an inlet and outlet water temperature difference of approximately 10.5℃@normal mode, 13.5℃@high hash mode
	Coolant Pressure	≤350kPa (Note: When the pressure is too high, it may cause the pipes, joints, water cooling plates to deform, causing the risk of coolant leakage)
	Coolant Medium	Special coolant: pure water (or distilled water) + special corrosion inhibitor + antifreeze (select the ratio according to the freezing point); Note: (1) The coolant must meet the index requirements listed in Table 3; (2) The coolant needs to be tested regularly. The test index and test cycle can be referred to in Table 4. When the test data exceeds or falls below the test index, its performance will not meet the requirements and the coolant must be replaced; (3) It is recommended to replace the coolant after one year of use.

	Coolant Circulation System	Circulation System: The system should be rust-resistant and corrosion-resistant. Stainless steel pipelines are recommended, while copper is prohibited. Filtration: The liquid medium's particle diameter should be ≤149 microns. Therefore, the main circulation path should include a filter with a mesh size of more than 100. Additionally, a 10-micron side filter is recommended to capture tiny suspended particles. System Cleaning: Before connecting the cabinet to the cooling system, clean and filter the system pipelines multiple times using deionized water (conductivity ≤ 5 μS/cm) to remove impurities like dust and welding slag. The cleaning process should not exceed 8 hours, and the cleaning water should not remain in the system overnight. After cleaning, ensure all deionized water is completely drained before injecting the coolant for circulation. Residual deionized water may affect the coolant's performance. Temperature Tolerance: System components must be capable of withstanding temperatures above 85°C. Safety Features: The system must include a 3.5 bar safety pressure relief valve. Pressure Management: The system should be equipped with a fixed pressure expansion tank. Note: As the coolant temperature rises after the server is powered on, it will cause an increase in system pressure.

	Humidity	1. Working humidity: 5%RH~85%RH (no condensation); 2. Storage humidity: 5%RH~90%RH (no condensation);
	Operation Ambient Temperature	Please ensure the device's operating ambient temperature is lower than the liquid inlet temperature (ambient temperature not greater than 50℃).

Note：The above-mentioned coolant temperature and flow parameters are the requirements for pure water + special corrosion inhibitor as the coolant medium. If the coolant medium comprises pure water + special corrosion inhibitor + antifreeze, its temperature and flow parameters are shown in Table 2.

Table 2 Recommended Temperature and Flow Parameters at Different Coolant Concentrations.

Coolant Medium	20% ethylene glycol or propylene glycol + special corrosion inhibitor	30% ethylene glycol or propylene glycol + special corrosion inhibitor	40% ethylene glycol or propylene glycol + special corrosion inhibitor	50% ethylene glycol or propylene glycol + special corrosion inhibitor
Freezing Point	-8.8℃ (20% ethylene glycol); -7.4℃ (20% propylene glycol);	-15.7℃ (30% ethylene glycol); -13.2℃ (30% propylene glycol);	-24.8℃ (40% ethylene glycol); -21.5℃ (40% propylene glycol);	-37℃ (50% ethylene glycol); -33.6℃ (50% propylene glycol);

Coolant Temperature	Inlet temperature: 20℃~55℃@normal mode; 20℃~44℃@high hash mode; Inlet temperature accuracy: ±2℃; Storage and transportation temperature: -30~70℃; Note: The coolant in the equipment must be drained if the storage and transportation is more than 2 hours.	Inlet temperature: 20℃~54℃@normal mode; 20℃~43.5℃@high hash mode; Inlet temperature accuracy: ±2℃; Storage and transportation temperature: -30~70℃; Note: The coolant in the equipment must be drained if the storage and transportation is more than 2 hours.	Inlet temperature: 20℃~53℃@normal mode; 20℃~43℃@high hash mode; Inlet temperature accuracy: ±2℃; Storage and transportation temperature: -30~70℃; Note: The coolant in the equipment must be drained if the storage and transportation is more than 2 hours.	Inlet temperature: 20℃~52℃@normal mode; 20℃~42.5℃@high hash mode; Inlet temperature accuracy: ±2℃; Storage and transportation temperature: -40~70℃; Note: The coolant in the equipment must be drained if the storage and transportation is more than 2 hours.
Coolant Flow Rate	Rated flow: ≥10.7L/min; Flow accuracy: ±10%; This flow corresponds to an inlet and outlet water temperature difference of approximately 10.5℃@normal mode, 13.5℃@high hash mode.	Rated flow: ≥11.0L/min; Flow accuracy: ±10%; This flow corresponds to an inlet and outlet water temperature difference of approximately 10.5℃@normal mode, 13.5℃@high hash mode.	Rated flow: ≥11.3L/min; Flow accuracy: ±10%; This flow corresponds to an inlet and outlet water temperature difference of approximately 10.5℃@normal mode, 13.5℃@high hash mode.	Rated flow: ≥11.7L/min; Flow accuracy: ±10%; This flow corresponds to an inlet and outlet water temperature difference of approximately 10.5℃@normal mode, 13.5℃@high hash mode.

Table 3 Initial index requirements for coolant

Details	Unit	Indicators

pH（20℃）	/	7.0~8.7
Conductivity (20℃)	μS/cm	TBD
Total colony count (microorganisms)	CFU/mL	<100
Sulfate	mg/L	<10
Chloride	mg/L	<20
Sulphide	mg/L	＜1
Total hardness (CaCO3)	mg/L	<1
Copper ion	mg/L	<0.5
Iron ion	mg/L	<0.5
Aluminium ion	mg/L	<0.5
Reserve alkalinity	mL	2.9~3
Appearance	/	Clear liquid without precipitation
Corrosion inhibitor	/	100% active ingredient

Table 4: Coolant Testing Index Requirements

Items (Note: Items Marked With * Are Mandatory Items)	Unit	Testing Indicators	Testing Period	Test Instruments/Methods
Ph（20℃）*		7.5~9.5	Every 2 months	PH metre/ASTM E70
Conductivity (20°c)*	μS/cm	Increment ≤1500	Every 2 months	Conductivity metre/GB/T11446.4
Total Colony Count (Microorganisms)*	CFU/mL	≤1000	Every 6 months	3M Bacteria culture dish SN/T 1897
Sulfate	mg/L	≤10	Every 6 months	Ion chromatography/HJ84
Chloride	mg/L	≤20	Every 6 months	Ion chromatography/HJ84
Sulphide	mg/L	≤1	Every 6 months	Ion chromatography/HJ84
Total Hardness (Caco3)	mg/L	≤20	Every 6 months	GB/T 6909
Copper Ion*	mg/L	Increment ≤ 0.1	Every 6 months	Inductively coupled plasma spectroscopy/HJ 776
Iron Ion*	mg/L	Increment ≤ 0.1	Every 6 months	Inductively coupled plasma spectroscopy/HJ 776
Aluminium Ion*	mg/L	Increment ≤ 0.1	Every 6 months	Inductively coupled plasma spectroscopy/HJ 776
Reserve Alkalinity	mL	≥2.3	Every 2 months	Automatic potentiometric titrator/ASTM D11221
Appearance*		Clear and transparent liquid without precipitation	Every 2 months	Visual Inspection
Corrosion Inhibitor*		≥ 80% of active ingredients	Every 2 months	Ultraviolet spectrophotometer, ion chromatograph, gas chromatography-mass spectrometer, etc.

3. Cooling system design requirements

3.1 Server Inlet Temperature Control

(1) Control requirements: constant inlet temperature control, control accuracy ±2℃.

(2) Control range: 20℃~55℃@normal mode (pure water + special corrosion inhibitor as coolant medium); 20℃~50℃@high hash mode (pure water + special corrosion inhibitor as coolant medium); other coolant temperatures refer to Table 2.

Note: Significant fluctuations in inlet temperature can lead to corresponding fluctuations in server temperature, potentially resulting in unstable server performance; If the liquid flow is too low, the outlet temperature may rise excessively, hindering the machine's normal functioning.

3.2 Server Inlet Flow Control

(1) Control requirements: constant inlet flow control, control accuracy ±10%.

(2) Control range: single machine flow ≥10.2L/min@pure water + special corrosion inhibitor as coolant medium, 10.2L/min corresponds to inlet and outlet temperature difference of 10.5℃@normal mode, 13.5℃@high hash mode; refer to Table 2 for the flow of other coolants.

Note: Excessive fluctuations in inlet flow can lead to significant variations in machine temperature, potentially causing unstable operation. If the liquid flow is too low, the outlet temperature may rise excessively, hindering the machine's normal functioning.

3.3 Server water inlet pressure

(1) Control range: Liquid pressure ≤ 350kPa;

Note: Excessive pressure may lead to deformation of pipes, joints, water-cooling plates, etc., causing the risk of coolant leakage, which may cause damage to the server, and the damaged server will not be covered by the warranty.

3.4 Server coolant medium

(1) Liquid Medium: special pure water (or distilled water) + special corrosion inhibitor + antifreeze (select the ratio according to the freezing point);

Note: Corrosion inhibitors must be added to the coolant, which must adhere to the specifications outlined in Table 3. If the coolant does not meet these standards, it must be replaced promptly to prevent rust and corrosion of system components. Failure to do so may result in the cold plate or joints becoming corroded or blocked, potentially damaging the server.

(2) Regular testing and replacement: The coolant needs to be tested regularly. Please refer to Table 4 for testing cycle and index requirements. If it does not meet the standards, it must be replaced immediately.

3.5 Circulation cooling system

(3) Pipeline: The circulation system should be rust-proof and anti-corrosive. It is recommended to use stainless steel pipes and copper is prohibited in the system.

Note: The cold plate is composed of aluminium. If copper is present in the system, electrochemical corrosion may occur. Additionally, carbon steel pipe welds can contain significant amounts of welding slag, which is difficult to clean and prone to rust and corrosion. This increases the risk of corrosion and blockage in the cold plate or joints, potentially causing damage to the server. Such damage will not be covered under the warranty.

(4) Filter: The main circuit is equipped with a filter with a mesh size of 100 or more; it is also recommended to configure a 10-micron side filter system to filter tiny suspended solids.

Note: If the filter mesh size is too small, larger particles of impurities in the system may not be effectively filtered, leading to potential blockages in the cold plate or joints. This could result in server damage, which would not be covered under the warranty.

(5) Circulation pump: It is recommended to use a stainless steel pump. To improve system reliability, it is recommended to ensure two pipes on hand, one for use and one as a reserve.

Note: Cast iron pumps are prone to rust and corrosion, causing the risk of corrosion and blockage of cold plates or joints, which will further cause damage to the server, which would not be covered under the warranty.

(6) System pipeline pressure testing and leak detection: It is recommended to use an air compressor for gas inspections of the pipeline, and apply the soap bubble method to check the seal of the pipeline welds.

Note: If conducting a water inspection during winter, the system equipment

must be drained to prevent freezing, as equipment that cannot be thoroughly drained may suffer damage. Gas inspections are more efficient than water inspections, as they do not require draining or antifreeze measures during winter leak detection.

(7) Exhausting and draining valves: Install automatic exhaust valves at local high positions and install drain valves at local low positions. It is recommended to install manual exhaust valves in multiple places in the system to improve the first injection efficiency of the system.

(8) Safety pressure relief valve: It is recommended that the system be equipped with a 3.5 bar safety pressure relief valve to prevent the system pressure from being too high.

Note: Ensure the pressure relief valve is properly adjusted before starting the machine. If the system fails to relieve pressure and the server's water inlet pressure exceeds 350kPa, it may result in server damage. Damage caused in this manner is not covered under the warranty.

(9) Constant pressure expansion tank: It is recommended that the system be equipped with a constant pressure expansion tank to prevent system pressure fluctuations;

Note: Please make sure that the expansion tank pressure value has been adjusted before operation. If the pressure tank fails to work, this may cause the server water inlet pressure to be greater than 350kPa, potentially damaging the server. Damage caused in this manner is not covered under the warranty.

(10) Temperature resistance of system components: ≥85℃;

Note: If the temperature resistance of the cooling system components is below 85°C, it may lead to damage or leakage of liquid, potentially causing server damage. Such damage is not covered by the warranty.

(11) Cleaning system: Before installing the server in the shelves, the shelf system needs to be filtered and cleaned;

Note:

a. Prior to connecting the cabinet to the cooling system, clean the system pipes multiple times with deionized water (conductivity ≤ 5μS/cm) to remove impurities such as welding slag and dust. Ensure that the cleaning process does not exceed 8 hours, and do not leave the cleaning water in the system overnight.

b. After cleaning, completely drain the deionized water before injecting the coolant for circulation. Failure to fully drain the deionized water can adversely affect the coolant's parameters.

c. Excessive residual impurities in the system may lead to corrosion and blockage of the cold plate or joints, potentially damaging the server. Damage resulting from such issues is not covered by the warranty.

(12) Cleaning the server: Before installing the server in the shelves, the server needs to be cleaned;

Note:

a. Prior to connecting the cabinet to the cooling system, clean the system pipes multiple times with deionized water (conductivity ≤ 5μS/cm) to remove impurities such as welding slag and dust. Ensure that the cleaning process does not exceed 8 hours, and do not leave the cleaning water in the system overnight.

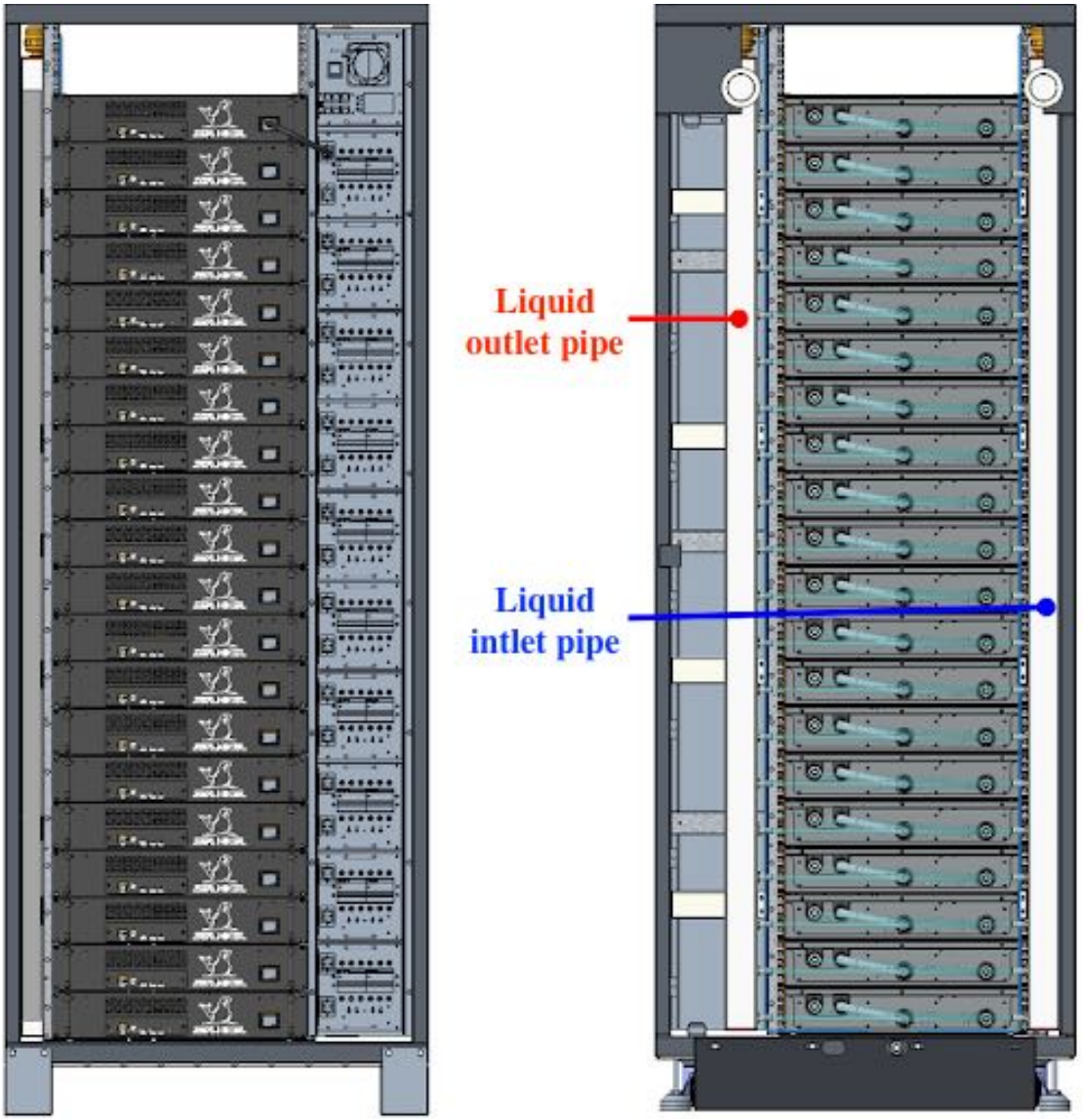
b. After cleaning, completely drain the deionized water before injecting the coolant for circulation. Failure to fully drain the deionized water can adversely affect the coolant's parameters.

C. Excessive residual impurities in the system may lead to corrosion and blockage of the cold plate or joints, potentially damaging the server. Damage resulting from such issues is not covered by the warranty.

4. Product installation and operation instructions

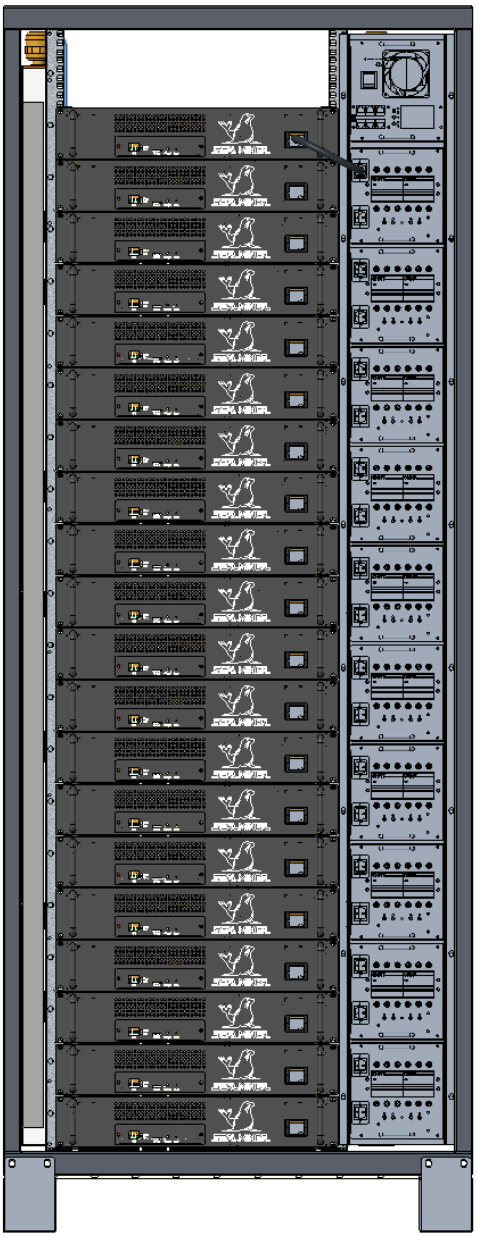
4.1 Installation location

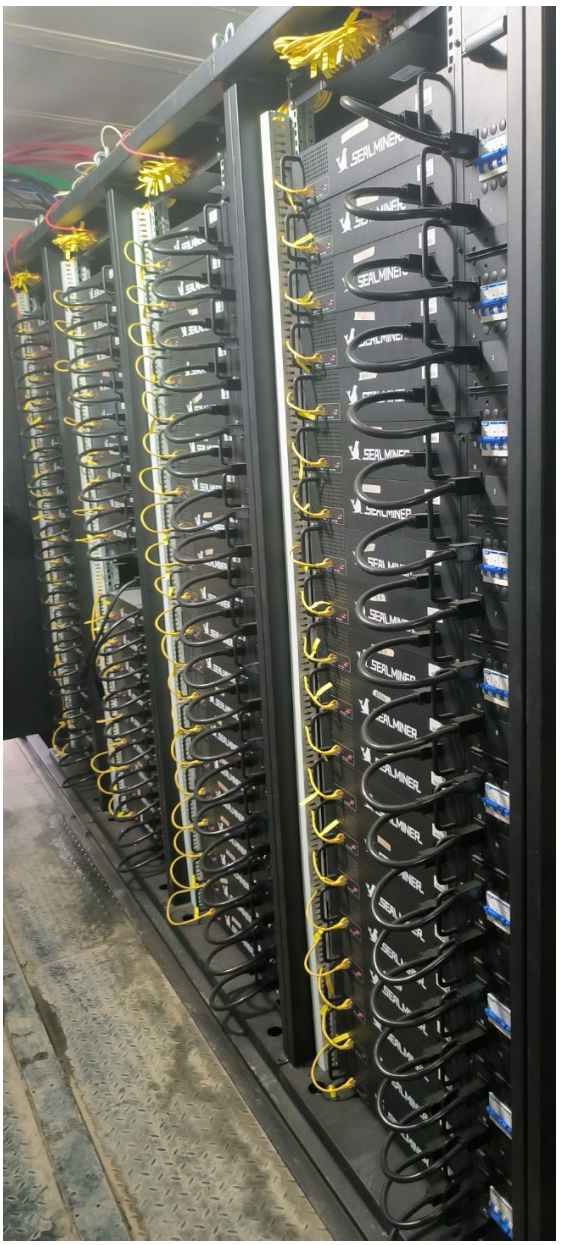
The water-cooled server adopts a standard 2U structure design and is installed in a standard 19-inch water-cooled cabinet. The front and back of the cabinet after the server is installed are shown in the following figure.



4.2 Installation Steps

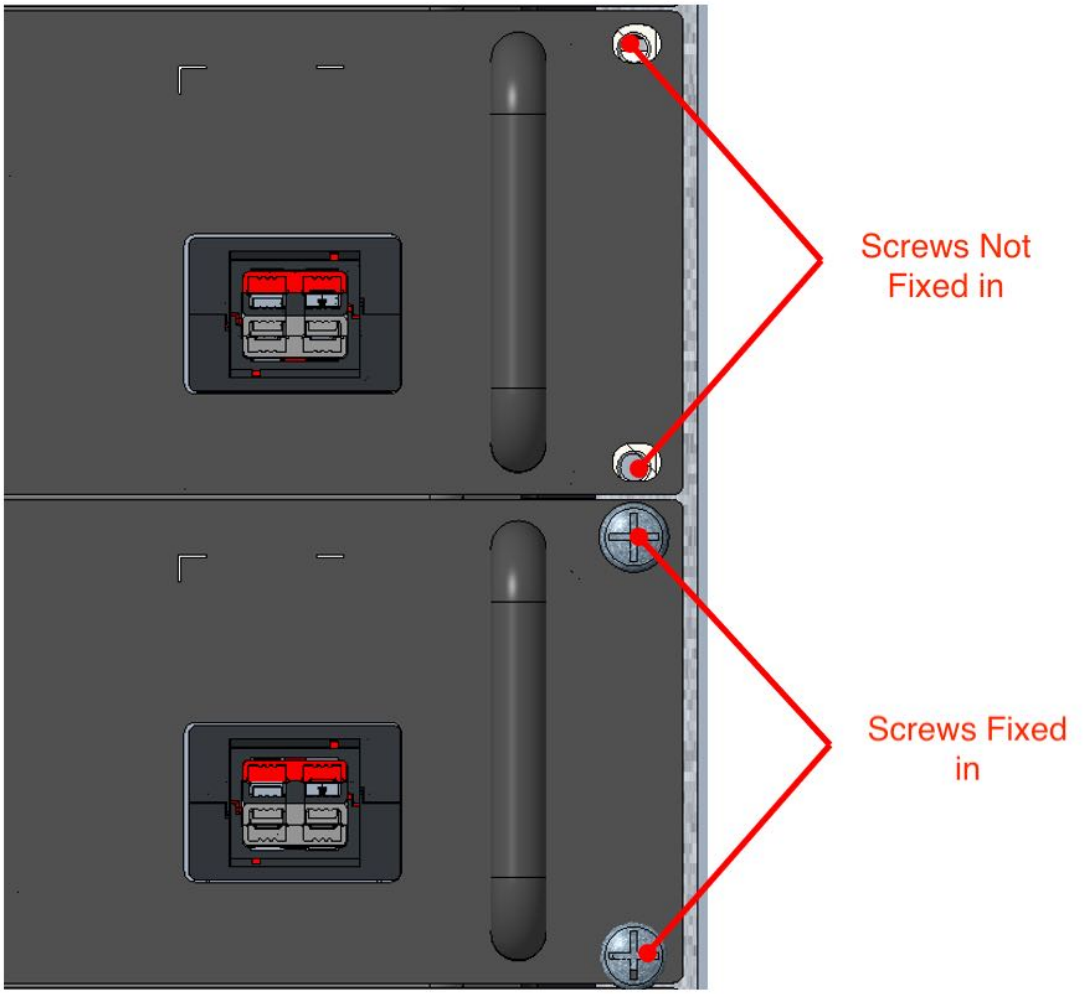
(1) Slowly push the server into the cabinet along the slot. It is recommended to have two people for the installation. A full cabinet with 20 servers is shown in the following figure.



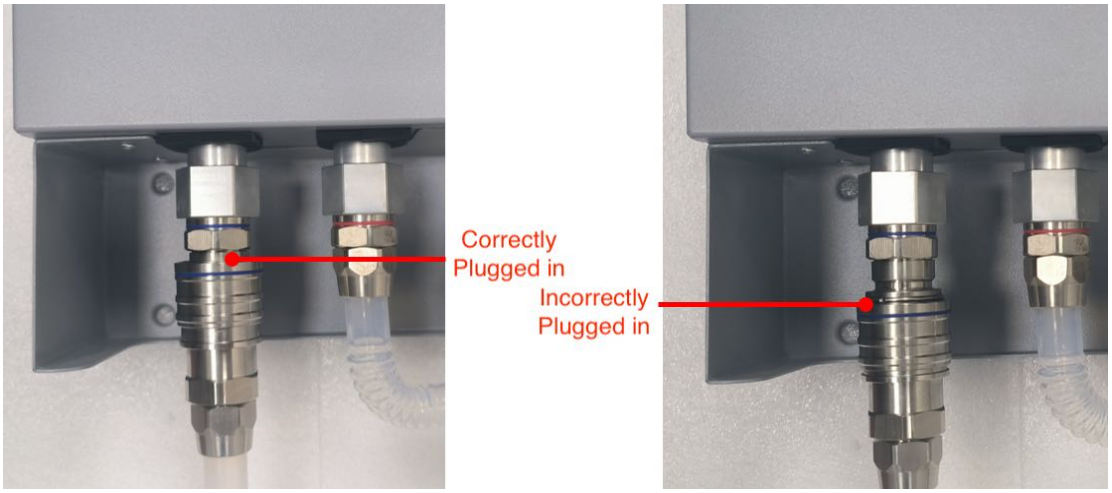


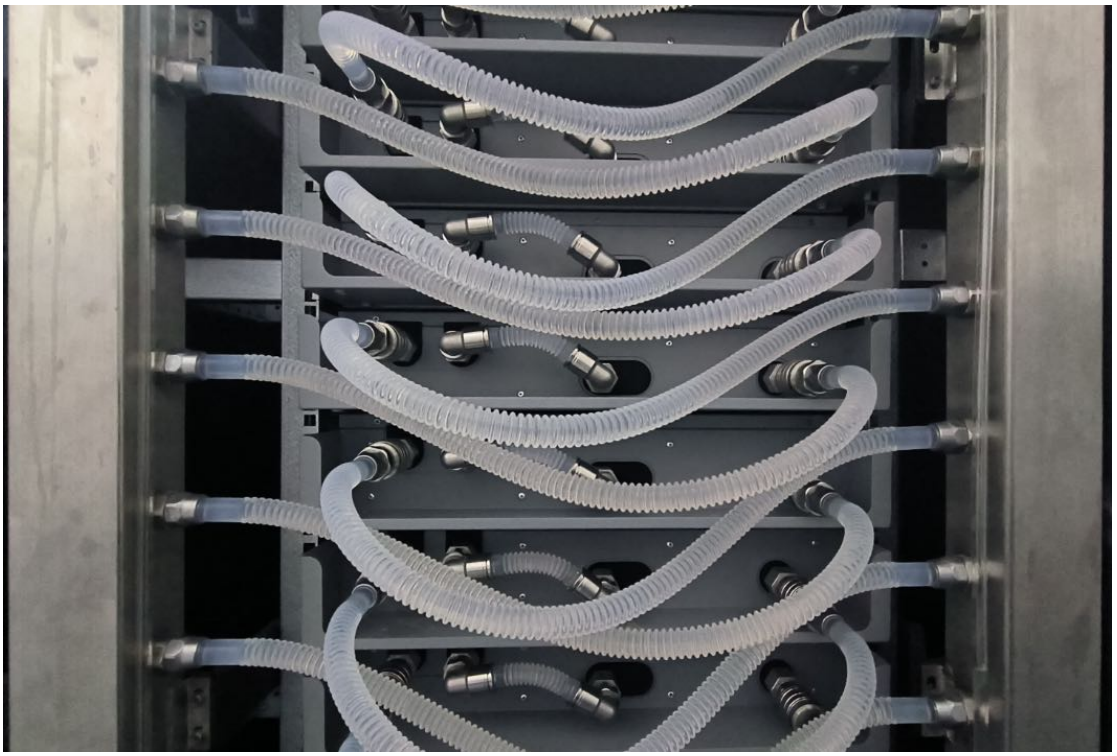
(2) Fix the server front panel to the cabinet with screws (the cabinet has floating nuts). The screws are fixed as shown in the figure below.

Note: It is not recommended to transport the servers within the cabinet. When transporting the servers within the cabinet, the front panel screws must be tightened and fixed. The servers damaged due to poor installation during transportation will not be covered by warranty.



(3) Plug the inlet and outlet pipes into the server. Start by removing the protective caps from the joints, then connect the female quick connector to the inlet and outlet pipes, ensuring they match with the male quick connector on the server. Make sure all connectors are securely in place. The accompanying figure illustrates this process: the upper part of the figure depicts two scenarios—one where the connectors are properly plugged in and one where they are not. The lower part of the figure shows an example of the correct setup after the connections are complete.





(4) Connect the network cable and power cable. The power cable uses the original factory configuration power cable. The example of the network cable and power cable connection is shown in the figure below.

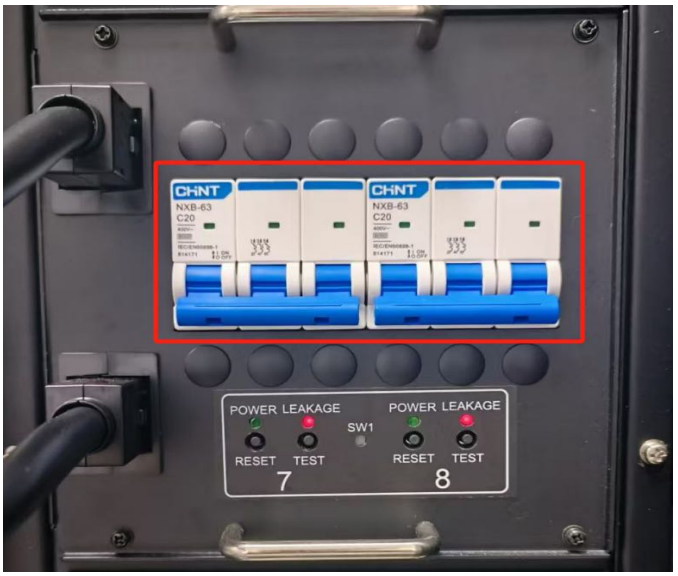


(5) After the server is installed, first circulate the water through the system to check for leaks. If there is no leakage, connect to the network and turn on the power to start running the servers.

Note: When filling the circulation system with the water, fill the system completely until no air remains within the system before powering on, otherwise there may be risks such as overheating, burning or computing power loss in the server, and the damaged machines are not covered by warranty.

4.3 Uninstallation Steps

(1) Shut down the servers and disconnect the corresponding circuit breaker, as shown in the figure below.

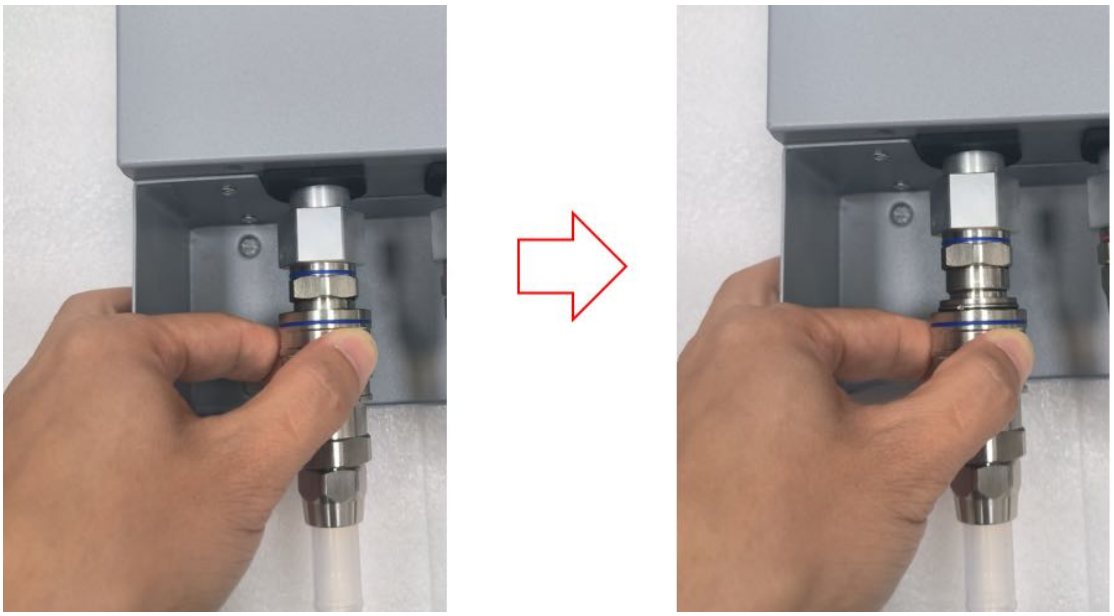


(2) Unplug the network cable and power cable, as shown in the figure below;

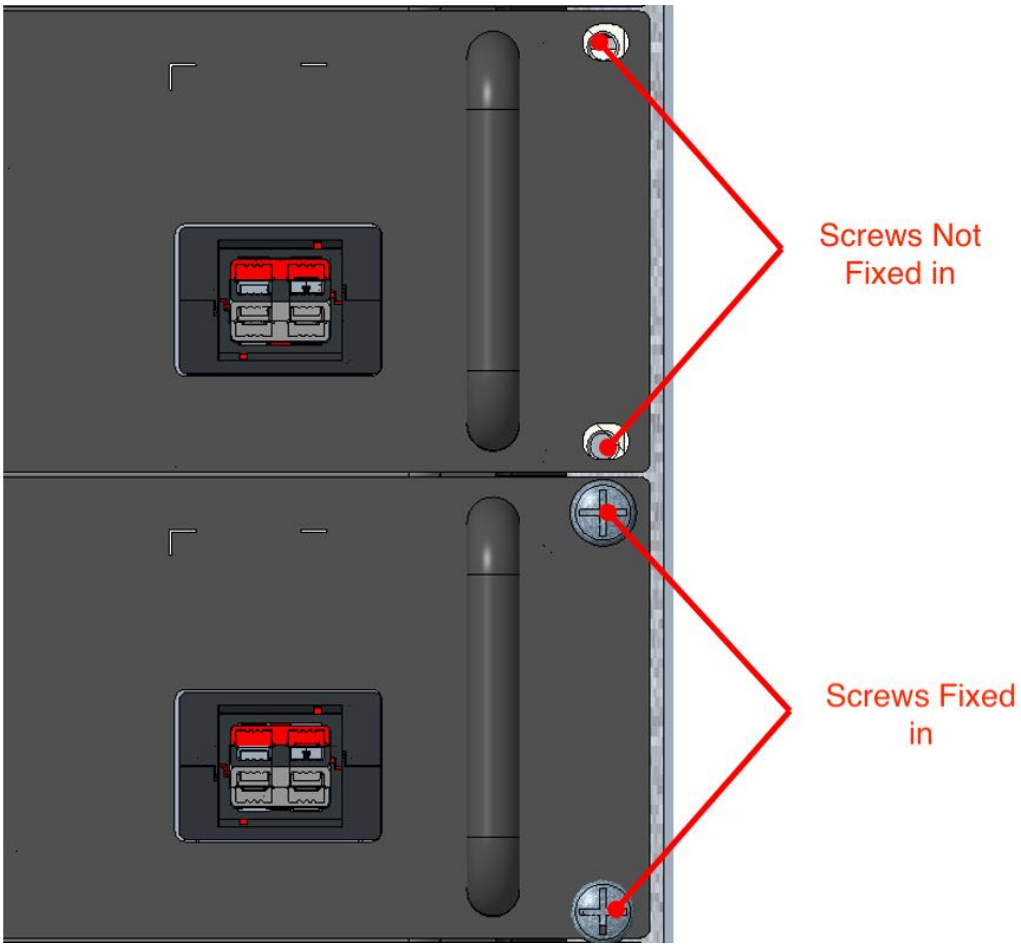


(3) Disconnect the water inlet and outlet quick connectors at the rear of the server. As shown in the figure below, hold the movable ring in the middle of the quick connector (female connector) and pull it back to disconnect.

Note: Disconnect the water inlet connector first and then the water outlet connector.



(4) Loosen the screws that fix the server front panel to the cabinet (the cabinet has floating nuts), as shown in the figure below.

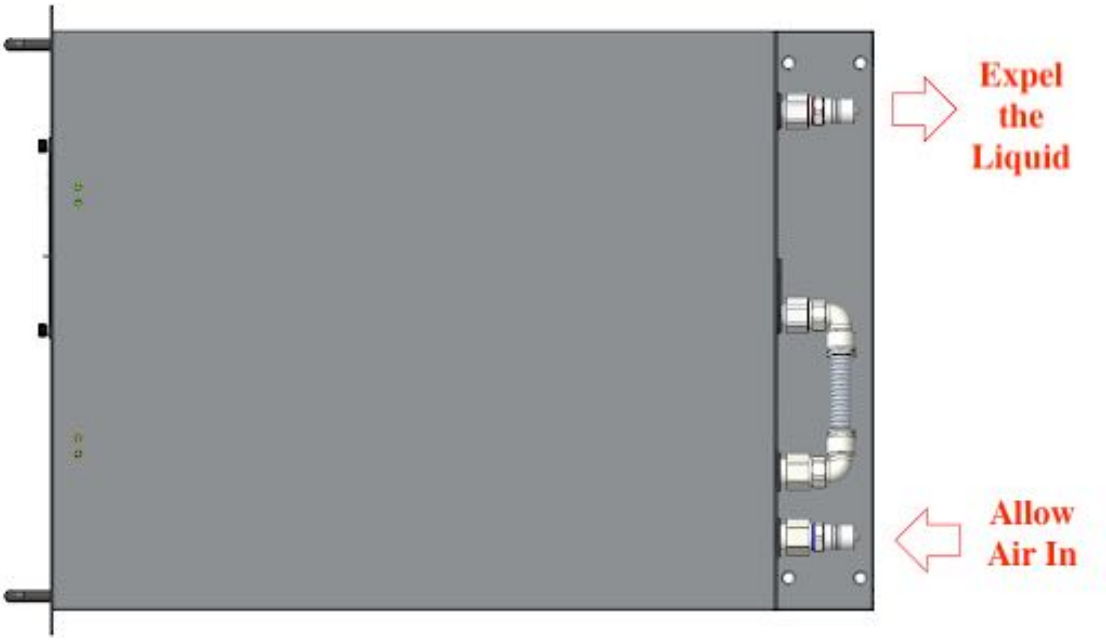


(5) To remove the server slowly pull the server out, it is recommended to have two people when removing the server , one on each side, as shown in the figure below;



(6) Drain the server thoroughly. After removing the server from the rack, if it will be stored or transported for more than 2 hours OR if the ambient temperature is lower than the freezing point of the coolant, ensure that all liquid inside the server is drained.

To drain the liquid inside the server, use two female quick connectors and connect them to two sections of water pipes, and then connect them to the water inlet and outlet of the server respectively. Fill one of the interfaces with air, the air pressure will discharge all remaining coolant, as shown in the figure below.



4.4 Note

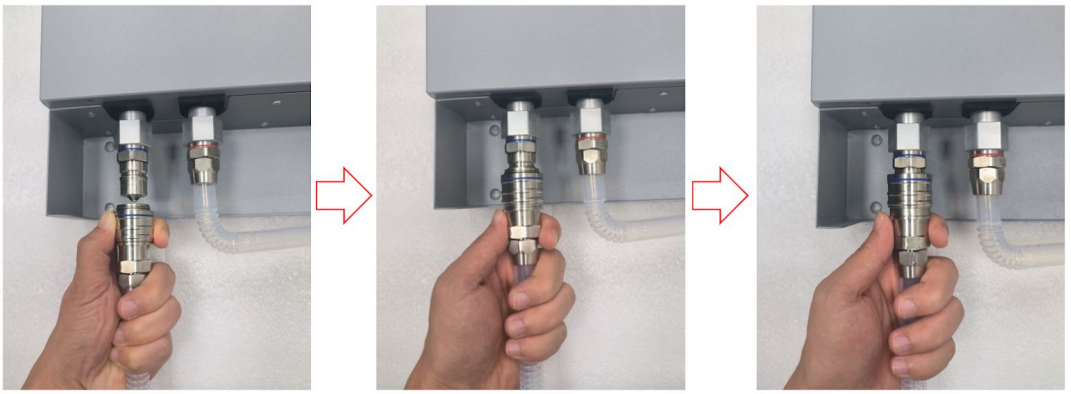
(1) Only original quick connectors and power cords should be used. Any

damage resulting from the use of non-original parts will void the warranty.

(2) The quick connector should be inserted and removed as demonstrated in the figure below. Upon proper connection, the male and female ends will align, allowing fluid to flow, and the quick connector will automatically lock. When disconnected, both ends will seal off, preventing leakage. Once the connection is made, you can verify that the quick connector is secure by gently pulling the hose at the female end to ensure it is not loose

Note: If the socket is not properly inserted, the server may overheat and become inoperable, potentially causing damage to the water cooling plate.

Insertion Instructions: Grip the movable ring on the female quick connector and pull it back, while simultaneously pushing the entire quick connector forward until it is securely in place. The movable ring will automatically return to its original position, indicating that the connection has been properly made.



1. Hold the base of the connector and pull the movable ring back

2. Insert the entire connector forward

3. After the connector is inserted forward, the movable ring will automatically return to its original position

(3) If the internal pressure of the server is too high due to improper power-on and power-off operations, the quick connector will not be able to be plugged in due to the high internal pressure of the server. In this case, as seen in the figure below, use a tool to press the active part of the water inlet or outlet quick connector to release the pressure inside the server.



(4) When disconnecting the power cord, press the springs on both sides of the connector, as seen in the figure below.

5. Important Notes for Powering On and Off the Product

5.1 Pre-Startup Checklist

(1) Cooling System Inspection: Before powering on the server, ensure that the entire cooling circulation system is functioning correctly, that any air in the pipes has been fully purged, and that there are no leaks in the system.

Note: Air in the system can reduce cooling efficiency and may cause damage to the circulation pump. Any leaks or micro-seepage in the system can lead to poor heat dissipation in nearby servers, increasing the risk of damage and negatively impacting the long-term performance of the cooling system. Special attention should be given to inspecting pipe connections, the server's water inlet and outlet fittings, and the intermediate connecting pipes. Damage caused by these issues will not be covered under warranty.

(2) Check Water Filling Status: Before placing the server on the shelf and powering it on, ensure that air has been purged from the server through water circulation and that the water cooling plate is fully filled. It is recommended to place one server on the shelf at a time, moving on to the next only after water is fully filled.

Note: Prior to installation, the water cooling plate of the server typically does not contain water, requiring connection to the system's circulating water. During this process, air will be introduced into the system, which may cause a temporary shortage of water in the pipe system. For this reason, the pipe system must be equipped with both an exhaust mechanism and an automatic liquid filling device. If multiple servers without coolant are connected in quick succession, the system may experience a temporary water shortage, preventing timely air expulsion. This can lead to risks such as circulation pump failure, overheating, or damage to local chips in other servers, and reduced computing power. Damage caused by these issues will not be covered under warranty.

(3) Check for Condensation Before Startup

Note: When the server is idle, a rapid increase in ambient temperature may cause condensation if the server cannot acclimate to the temperature change. In the event of condensation, the server can be dried using external heating methods, such as sunlight or hot water. Any damage resulting from condensation will not be covered under the warranty.

5.2 Startup Procedure

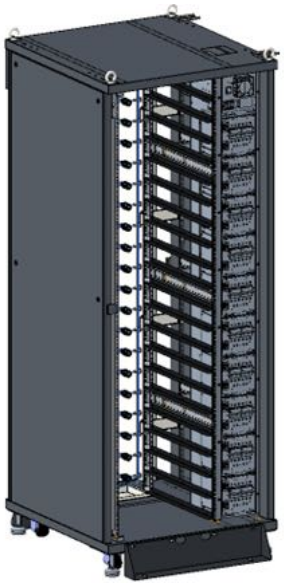
When powering on all servers simultaneously within the same cooling system, to prevent the water temperature control from becoming overwhelmed and triggering high-temperature protection, it is advisable to fully activate the cooling system's capacity before starting all servers. This means that all temperature control components, such as fans, circulation pumps, and electric valves, should be operating at full capacity. Once the servers approach their rated power levels, the cooling system will automatically adjust to maintain optimal water temperature.

5.3 Shutdown Precautions

After the server is shut down and its water inlet and outlet are disconnected from the cooling system (i.e., when the server is no longer circulating water), it is essential to drain the cooling water from the server to prevent damage to the cold plate.

Note: Any damage resulting from failure to drain the cooling water will not be covered under the warranty.

6. Product Accessory Specifications

Item	Specification	Image for Reference	Remarks
Water Cooling Cabinet	Effective Space: 43u Dimensions (Height X Width X Depth): 2100*800*850mm Water Supply And Return Side Interface Size: M16*1.0 Screw Thread		The cabinet integrates switches, water supply and return manifolds and PDUs, allowing 20 2U servers to work simultaneously
Quick Connect or (Male)	Interface Size: M16x1.0 Screw Thread; Interface Sealing Method: Sealing Ring End Face Seal; Colour Identification: Blue		This connector is used for the server's original water inlet interface.
Quick Connect or (Male)	Interface Size: M16x1.0 Screw Thread; Interface Sealing Method: Sealing Ring End Face Seal; Colour Identification: Red		This connector is used for the server's original water outlet interface.

Elbow Connector	Interface size: M16X1.0 thread, matching FEP tube with outer diameter of 16mm; Interface sealing method: Sealing ring end face seal		This connector is used to connect the server's original power supply to the water cooling plate.
Quick Connector (Female)	Interface Size: Φ13 Pagoda Head, Matching Fep Tube With Inner Diameter 12mm And Outer Diameter 15mm; Colour Identification: Blue		This connector is designed to be used in conjunction with the server's original water inlet connector. The customer's cabinet water supply system must utilise this specific connector to ensure compatibility with our server's water inlet. Note: Customers must not use connectors of different specifications or brands to replace this connector, as forcing a mismatch with the original water inlet connector on the server may result in damage.

Quick Connector (Female)	Interface Size: Φ13 Pagoda Head, Matching Fep Tube With Inner Diameter 12mm And Outer Diameter 15mm; Colour Identification: Red		This connector is designed to be used in conjunction with the server's original water inlet connector. The customer's cabinet water supply system must utilise this specific connector to ensure compatibility with our server's water inlet. Note: Customers must not use connectors of different specifications or brands to replace this connector, as forcing a mismatch with the original water inlet connector on the server may result in damage.
Corrugated Tube	Interface Size: ϕ16*100mm		This hose is the original pipe used to connect the server's power supply to the water cooling plate.

Power Cord	L=360mm, Double-Ended With Plug, 4*2.5mm².		This power cord is used to connect the server power supply to the PDU on the server cabinet, with plugs at both ends designed to match the server's power interface (socket). **Note:** The PDU interface on the cabinet must be a socket that matches the plug on this power cord. This power cord is specifically designed for the server cabinet setup provided by our company.

7. Network Configuration

7.1 Server Configuration Device List

No.	Item	Qty	Usage	Remarks
1	Computer	1 Set	Server Configuration Operations	
2	Network Switches	1 Set	Configure servers and configure operations Computer network communications	The network switch is capable of connecting to the external network
3	Dhcp/Router	1 Set	Assign a dynamic IP address to the server upon initial power-up.	The default setting of the server is DHCP to obtain a dynamic IP address.

7.2 Server Network Environment

The factory default for the server is to obtain a dynamic IP through DHCP, so a DHCP server must be configured in the network, or the router must enable the DHCP dynamic IP address service.

8. Data Configuration

8.1 Query the Dynamic IP Address Assigned to the Server

Connect the computer to the same network segment as the server, use the Sealminer management tool to detect, and click "Detect" to open the IP detection window



Click "Start" on the window to start detection as seen in the figure below. (Note: The mining pool address, miner name, and password in the figure below are SEALMINER Demo examples. For actual operations, please refer to your own mining pool address, miner name, and password.)



Reporting IP method: After starting the mining machine, press and hold the IP Report button on the mining machine for 5 seconds as seen in the figure below. Wait for a few seconds or until the current device IP is displayed on the software (Note: the management tool PC and the mining machine must be in the same LAN).



Check the acquired IP and MAC addresses reported by the server in the Sealminer Tool software.

Note：

(1) If all lights on the server control panel are off after powering on, verify that the power cord is securely and correctly connected.

(2) If the indicator light on the right side of the server control panel is on, but the

network port light is off or the green light does not flash, check the functionality of the switch, ensure the network cable connection is secure, and confirm the quality of the network cable.

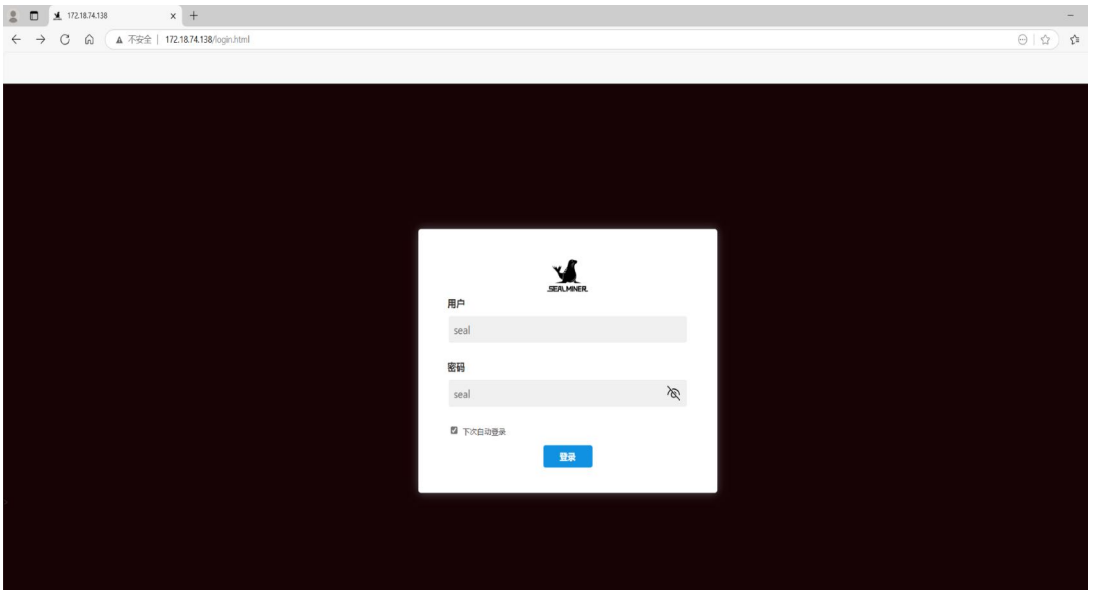
(3) The computer running the Sealminer Tool software and the server must be on the same network segment. Otherwise, the software may not receive the broadcast message sent by the server, preventing it from querying the IP address and MAC address information reported by the server's IP Report button.

(4) If the computer and server are in the same network segment and the DHCP service is enabled in the network, but the Sealminer management tool does not query the server IP after pressing the server IP Report button. Press and hold the Reset button on the server panel for more than 5s to restore the factory default configuration, then power off the server and restart it. After the server starts normally, press the IP Report button for 5s to detect the server IP address.

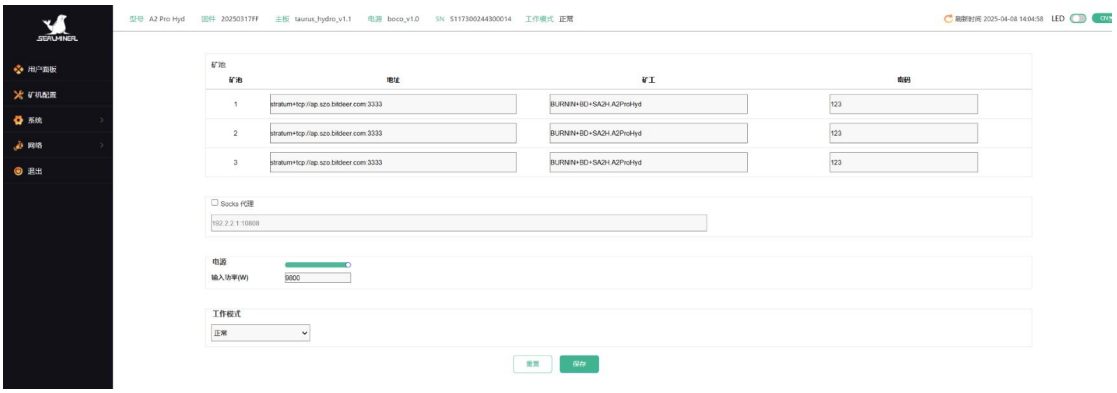
8.2 Mining pool & miner data

(1) Log into the webpage

Account: seal, Password: seal



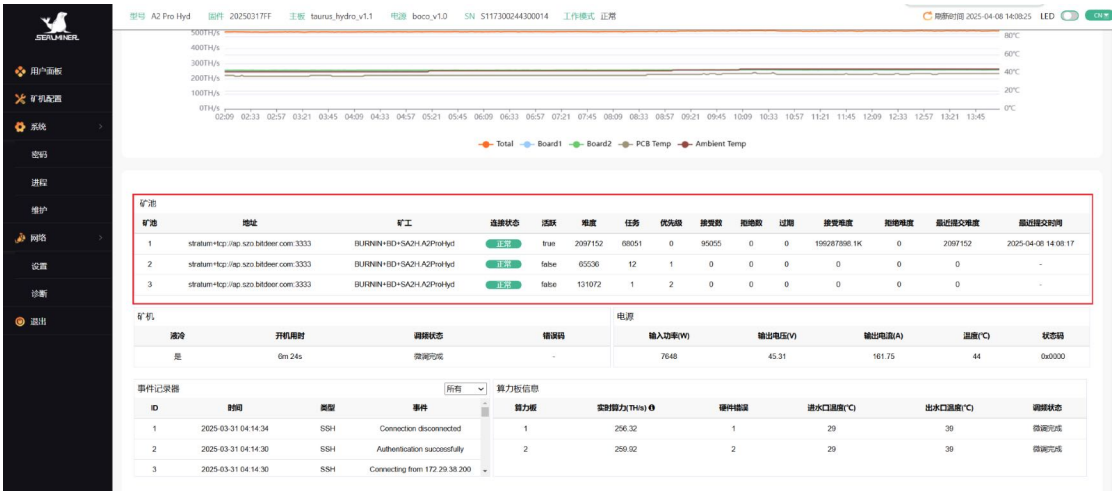
(2) In the mining machine configuration interface, modify the mining pool address and miner name. After making the changes, click "Save" below. (Note: The mining pool address, miner name, and password in the figure below are SEALMINER Demo examples. For actual operations, please refer to your own mining pool address, miner name, and password.)



Configuration Update Confirmation: Changes to the pool configuration will take effect immediately without requiring a restart of the Miner program or the mining machine.

(3) Verify Configuration Changes

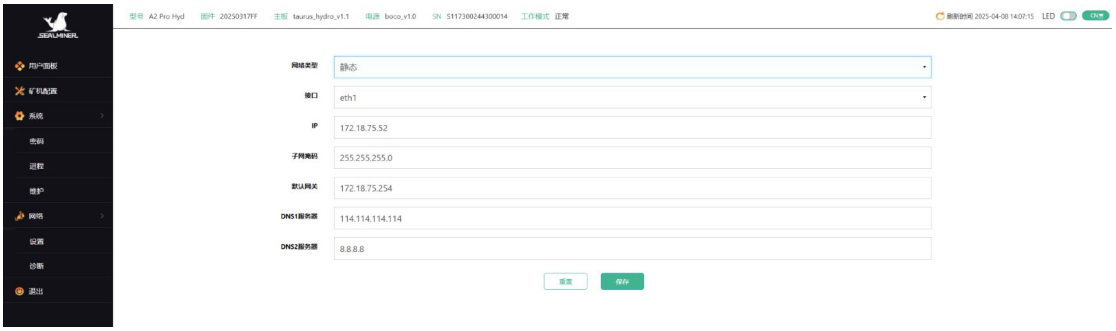
To ensure that the configuration modifications have been applied, go to the User Panel interface and verify that the updated mining pool and miner data are correctly reflected. (Note: The mining pool address, miner name, and password in the figure below are SEALMINER Demo examples. For actual operations, please refer to your own mining pool address, miner name, and password.)



8.3 Configuring a static IP address for the server

(1) Log into the web page

(2) On the Network->Settings page, select "Static" in the Network Type option, change the IP address, mask, gateway, and DNS address to the actual address planned for the mine, and click "Save".



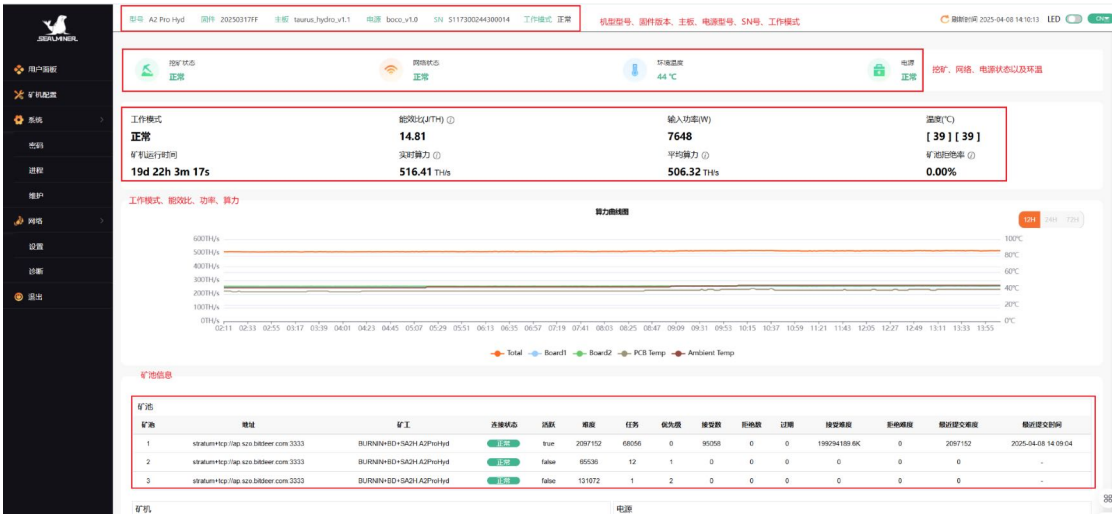
Note: After saving the configuration, use the newly set static IP address to log in to the server. Failure to do so will result in the page displaying "Loading" indefinitely until it fails to load.

9. Server Operation Status Check

After connecting the server to the network, log in to the server to review its operational status.

(1) In the server interface, select the "Overview" option to access the Miner running status page.

(2) Review the server's overall computing power, connected mining pool, individual board computing power, single board temperature, and other relevant performance metrics. (Note: The mining pool address, miner name, and password in the figure below are SEALMINER Demo examples. For actual operations, please refer to your own mining pool address, miner name, and password.)



10. Batch Configuration

The Sealminer Tool software can be used to perform server batch data configuration, status monitoring, and firmware upgrades. For detailed operations, please refer to the "Sealminer Tool Operation Guide".

11. Disassembly and installation

11.1 Removing the Control Board

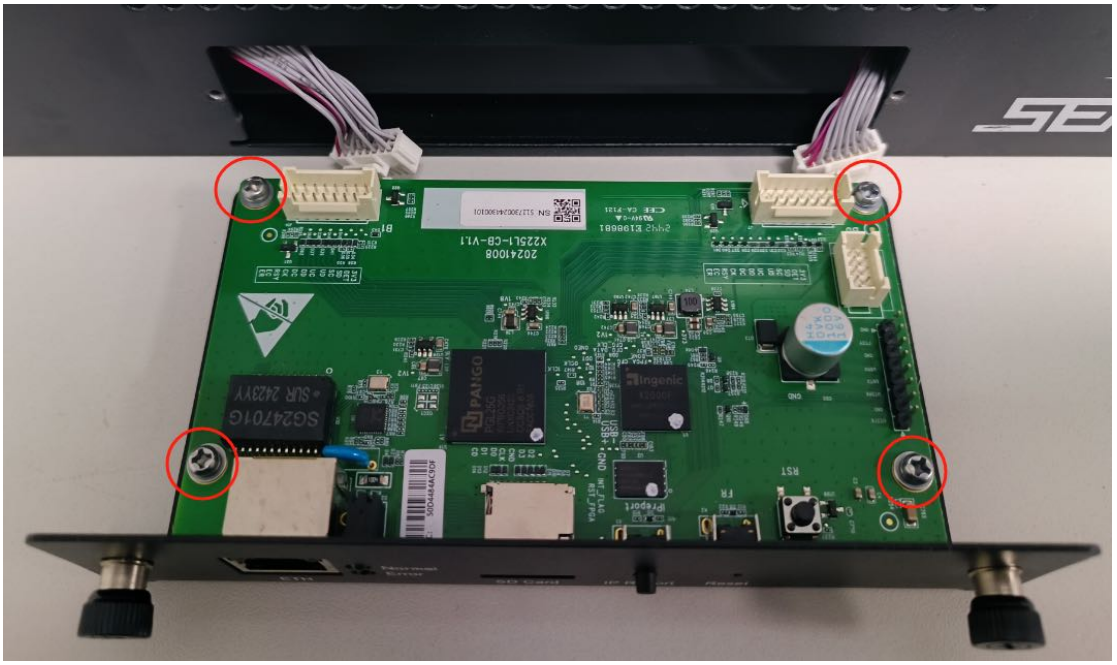
(1) Firstly, turn off the power, and then loosen the two cover screws on the control panel, and then pull out the board.



(2) Unplug the three cables as seen in the figure below.



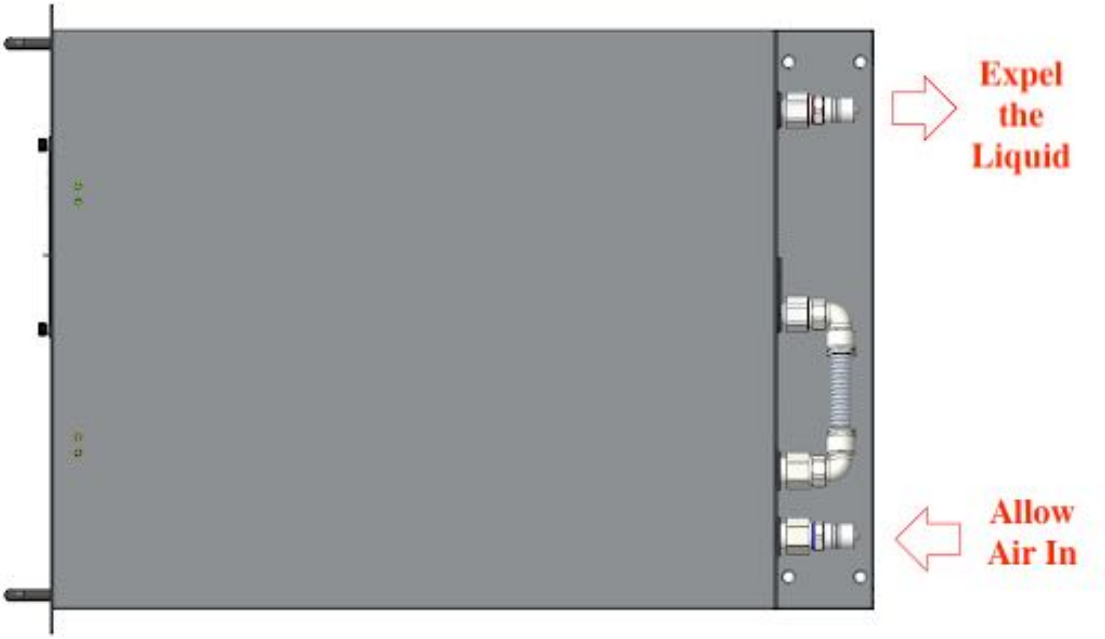
(3) Remove the 4 screws on the control board as seen in the figure below.



11.2 Disassembling the Hash Board

(1) First, power off the server and remove it from the cabinet.

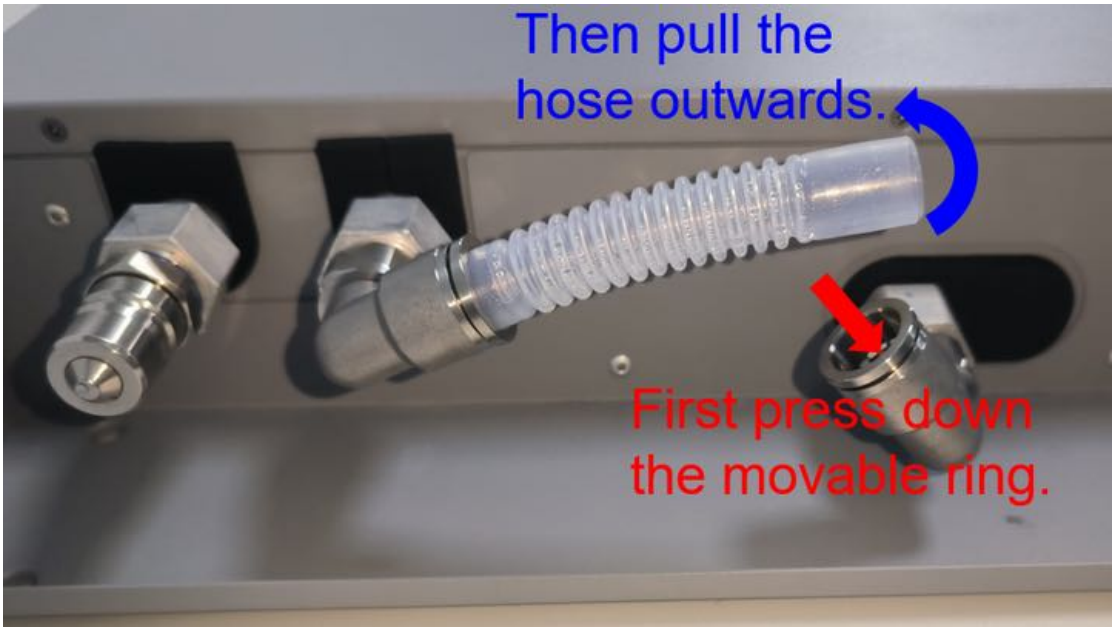
(2) Drain the coolant from the server, refer to the figure below or section 4.3 for instructions on draining the coolant.



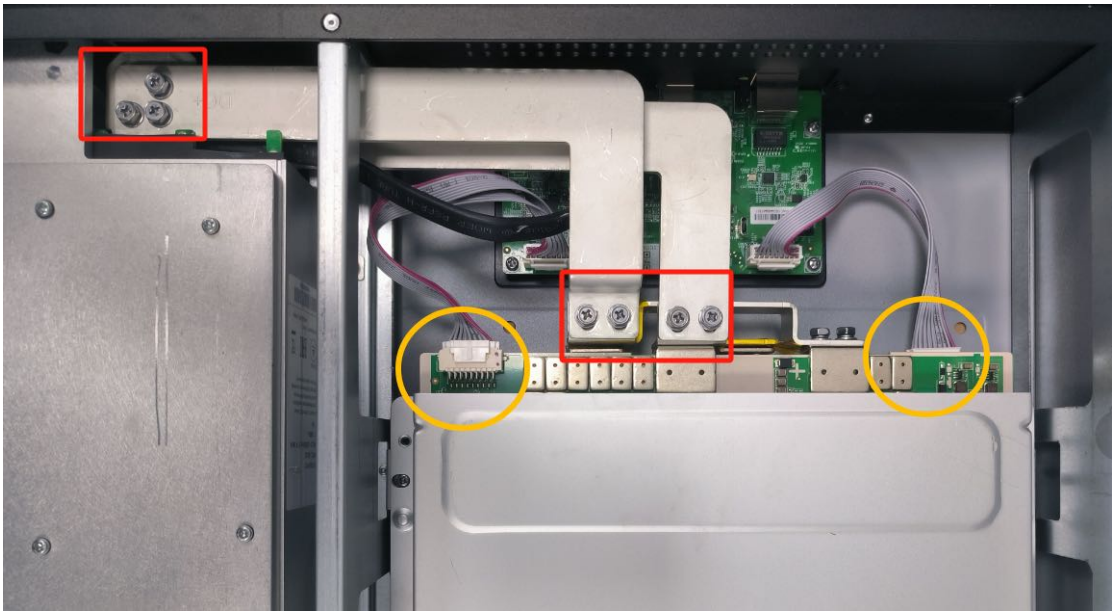
(3) Use a tool to press the water inlet or outlet quick-connect fitting to release the pressure inside the server.



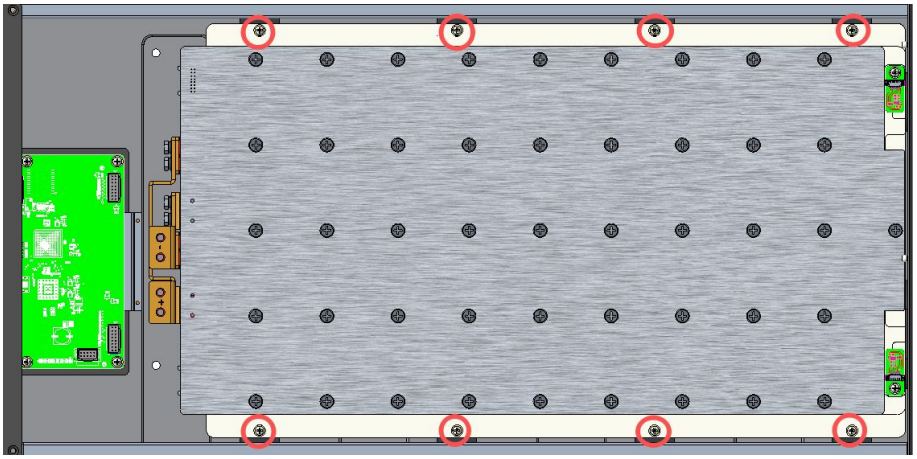
(4) Remove the corrugated hose as seen in the figure below. Press the movable ring of the quick-connect connector inwards, and then pull the corrugated hose out.



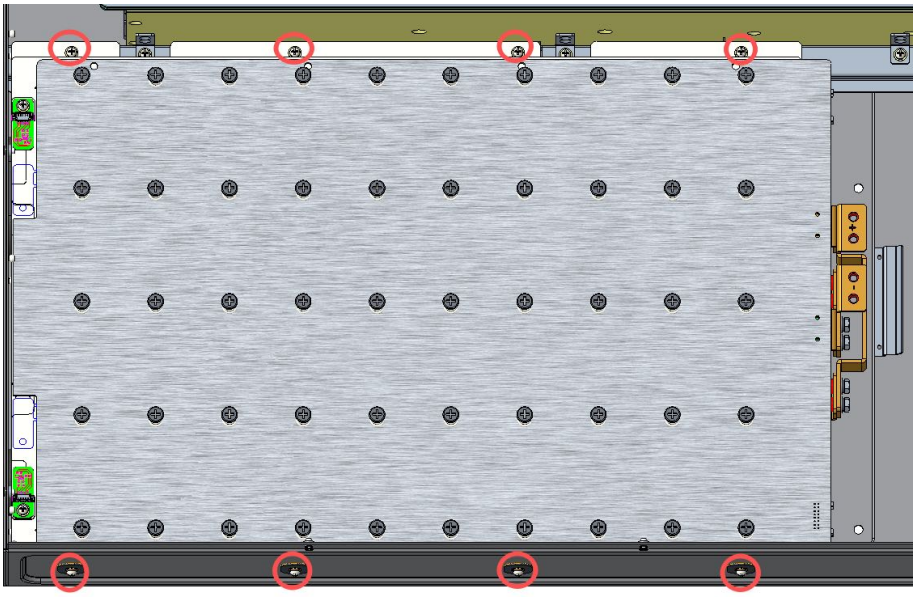
(5) After removing the server cover, remove the screws on the copper bus bars and cables connecting to the hash board.



(6) Remove the 8 screws on both ends of the water cooling plate to remove the entire water cooling plate.（Two architectures)

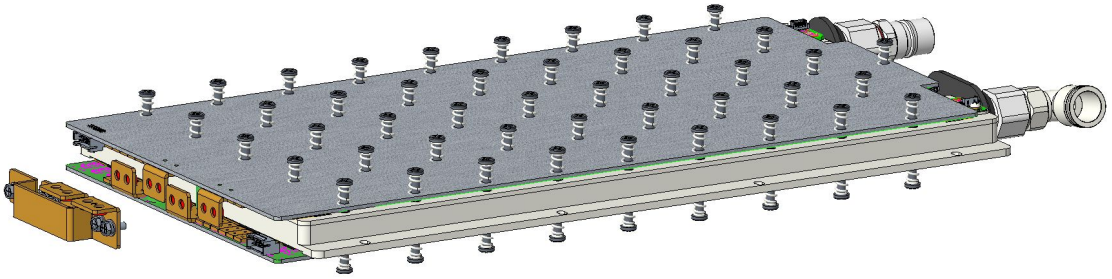


Architecture ①

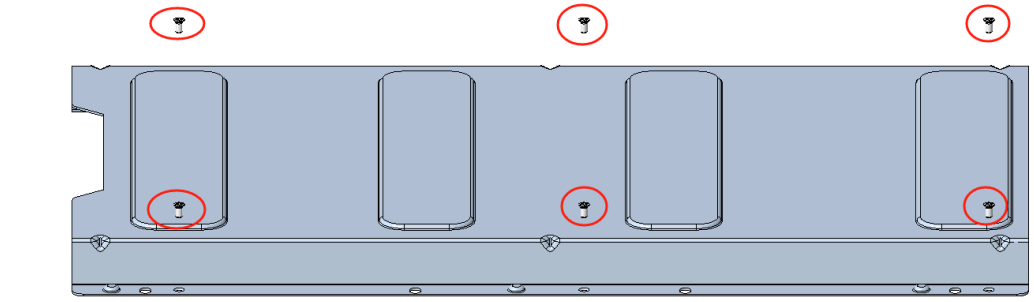


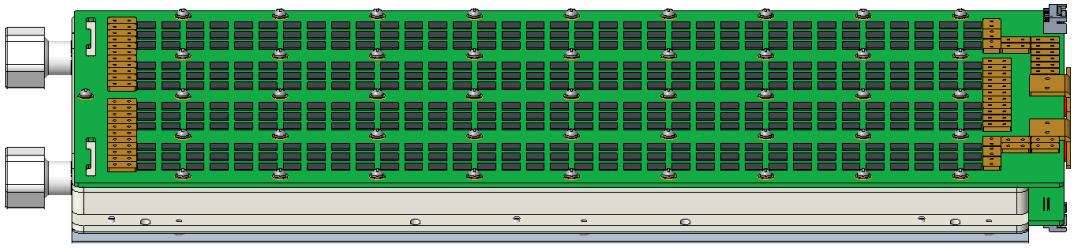
Architecture ②

(7) Remove the parallel copper busbar connecting the two hash boards.



(8) Remove the screws securing the protective cover of the hash board, then detach the cover.

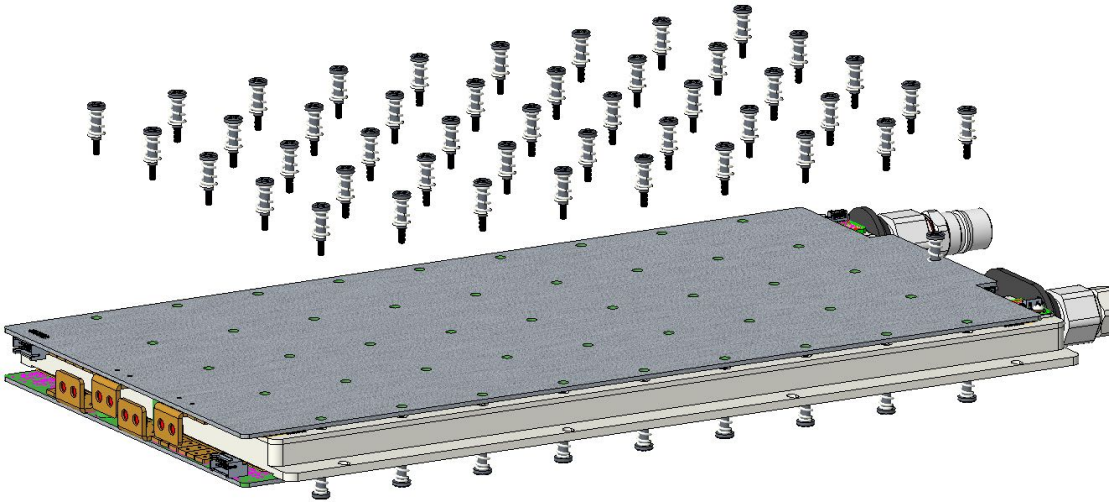




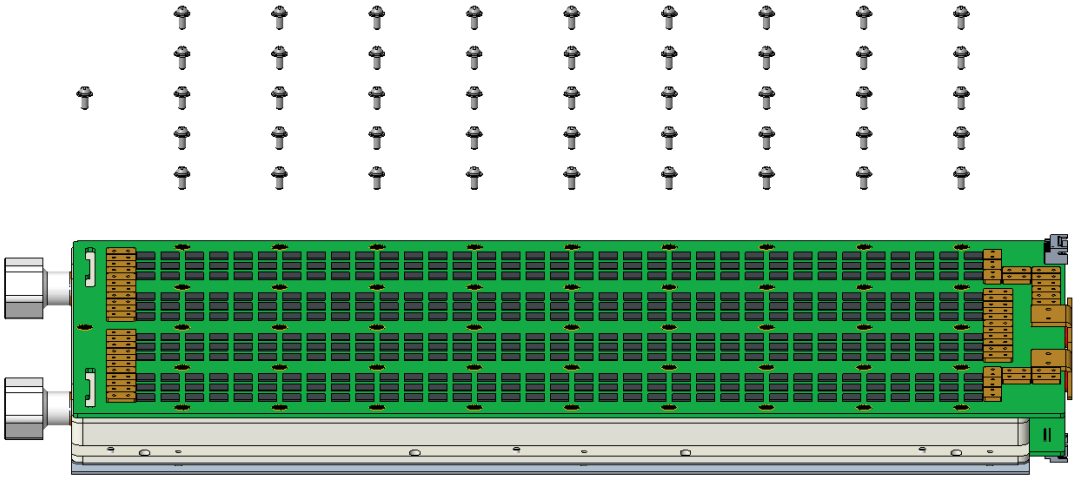
(9) Unplug the temperature sensor cables on both sides. (if applicable)



(10)Remove the screws securing the hash board, then lift out the hash board. (Two architectures)



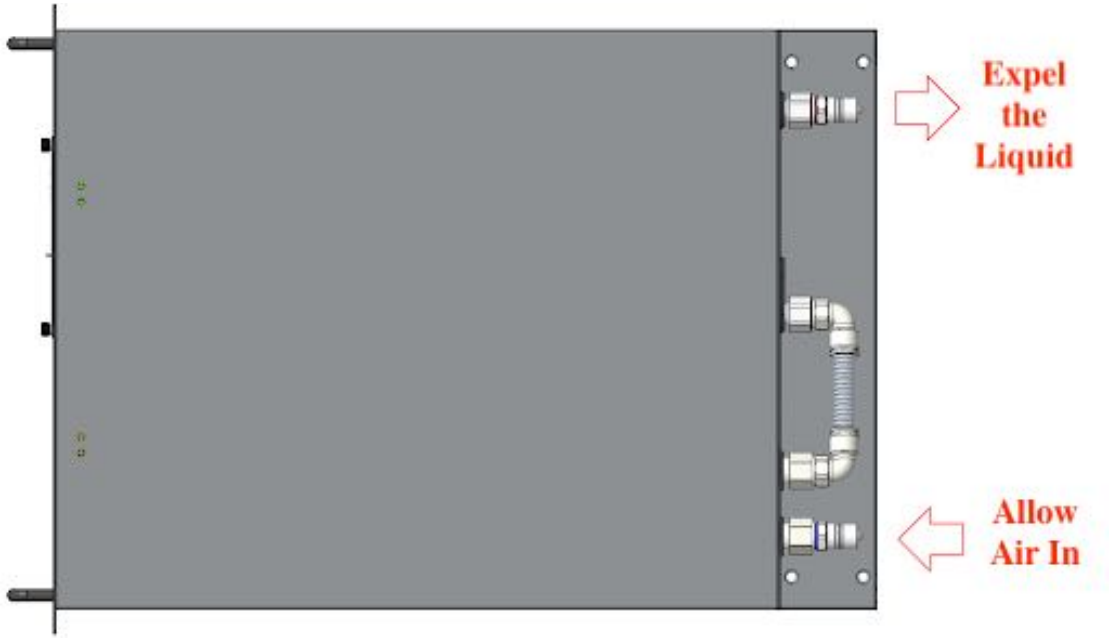
Architecture ①



Architecture ②

11.3 Removing the Power Supply

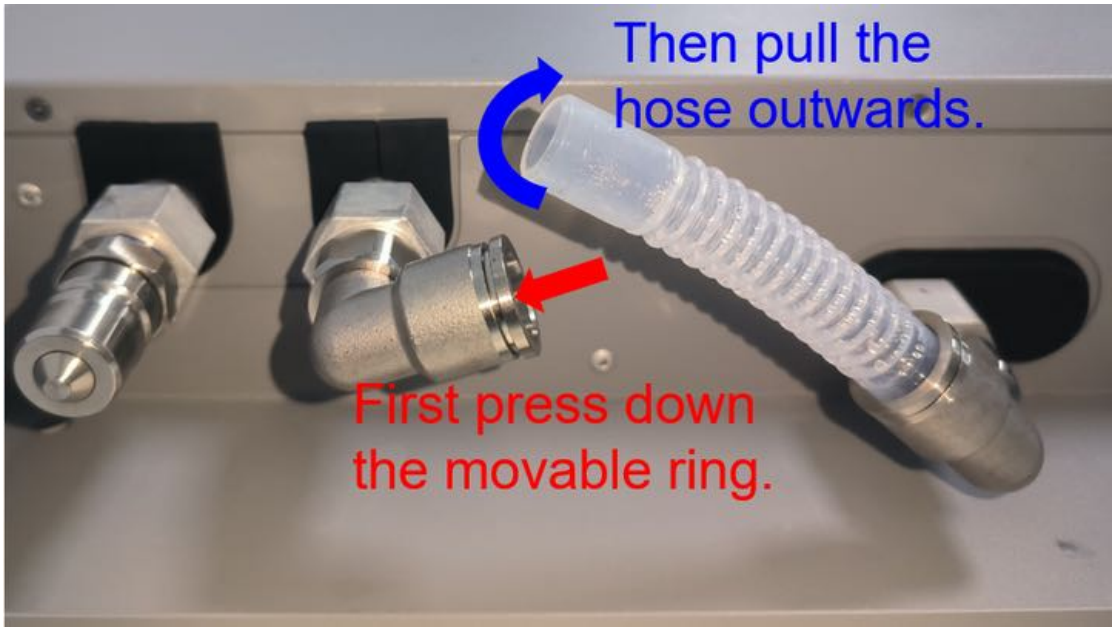
(1) Drain all of the coolant in the server.



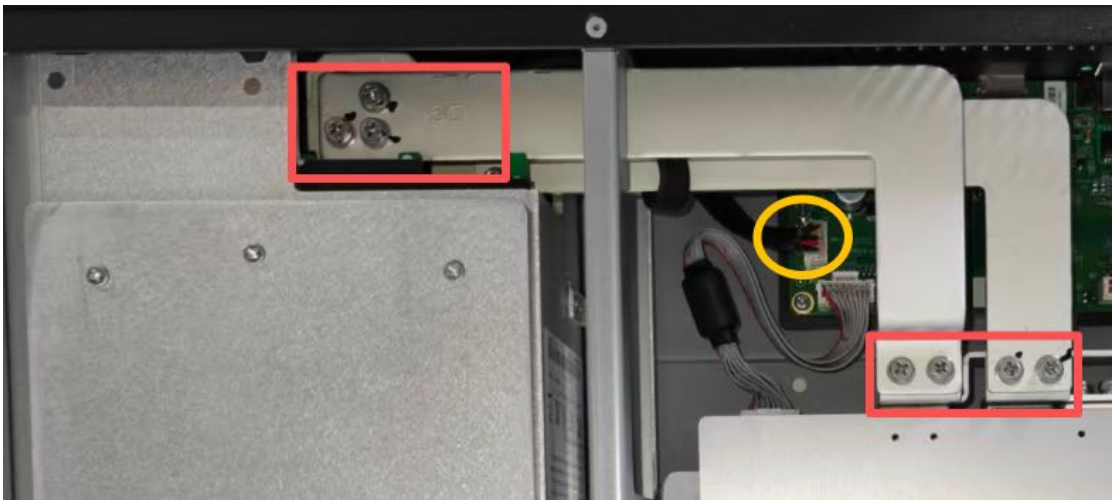
(2) Use a tool to press the water inlet or outlet quick-connect fitting to release the pressure inside the server.



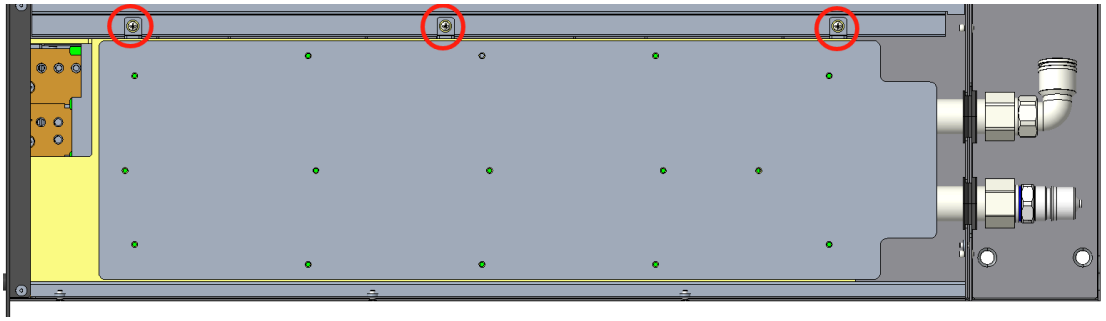
(3) Remove the corrugated hose as seen in the figure below. Press the movable ring of the quick-connect connector inwards, and then pull the corrugated hose out.

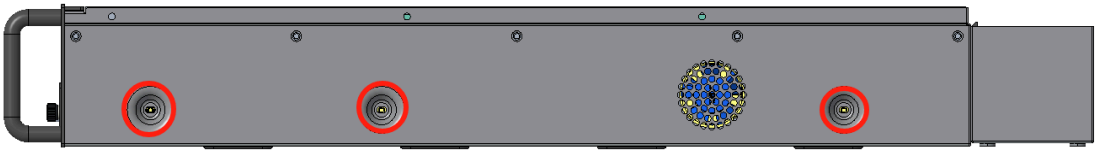


(4) After opening the server cover, remove the screws on the copper bus and unplug the cable as shown in the figure below.



(5) Remove the 6 screws on both sides of the power supply to remove the power supply as shown in the two figures below.





12. After-Sales Service

At Bitdeer, prioritising user experience is our primary objective. We actively seek customer feedback to enhance our services and are dedicated to assisting customers with their issues. Bitdeer offers a 1-year warranty on Hydro series products, including free repair services within the warranty coverage. By making a purchase, you agree to our warranty terms. However, please note that we will not provide repairs for the following situations or failures:

1. Damage resulting from failure to install, use, maintain, or service the machine according to the requirements outlined in the official documents.

2. Unauthorised disassembly, modification, assembly, or repair without official written or electronic authorization, leading to damage to the product.

3. Damage or loss caused by mishandling, misuse, abuse, improper operation, incorrect installation, or maintenance and storage that do not comply with regulations.

4. Product damage arising from operating the mining machine in environments that do not meet the specified requirements, including but not limited to high humidity, corrosive environments, electrical surges, extreme temperatures, abnormal voltage or current (including surges, impacts, and instability), excessively low or high AC voltage, excessive water inlet pressure, and improper use resulting in ice formation and deformation within the water-cooled product pipeline.

5. Damage to the entire machine, the panel, or components due to crushing, breaking, burning, or falling, resulting from improper operation or similar issues.

6. Product damage caused by overvoltage, undervoltage, or leakage.

7. Damage or loss resulting from unforeseen natural disasters, including but not limited to floods, fires, earthquakes, tsunamis, and lightning strikes.

8. Disassembly or modifications made by anyone other than us or our authorised service agencies.

9. Product failure or damage resulting from the use of accessories, parts, or components, such as power supplies, that are not manufactured by us or our authorised agencies and do not meet our specified parameters.

10. Failure or damage resulting from the use of unauthorised firmware or drivers, including but not limited to unauthorised overclocking firmware.

11. Altered, defaced, or removed SN labels.

12. Use of non-original or mixed boards: if some or all of the computing boards, control boards, or power supplies are not original to the machine or if there is any situation preventing us from verifying the authenticity of these components.

13. Situations not caused by our company that make it impossible for us to determine whether the product is still under warranty.

Due to the specific regulatory requirements for water-cooling products concerning system environment, media, machine voltage, temperature, flow rate, and other factors, in addition to the general warranty exclusions mentioned above, the following conditions also void our product warranty for water-cooling products:

1. Failure to equip the piping system with a constant pressure tank and a mechanical pressure relief valve. If the water inlet pressure exceeds 350 kPa, causing deformation of the cold plate and damage to the product.

2. Use of coolant that does not meet the specified requirements, leading to corrosion, blockage, or other damage to system components.

3. As the product cold plate is made of aluminium alloy, corrosion or blockage of the cold plate or joints is possible due to the presence of metals or materials prone to electrochemical corrosion (e.g., copper, cast iron, carbon steel) in system components.

4. Inadequate filter mesh causing particulate impurities to enter the system, leading to blockage of the cold plate or joints and resultant damage.

5. Damage to system components or leakage of the liquid medium if the cooling system components have a temperature resistance lower than 85°C.

6. Failure to thoroughly clean and filter the system pipelines with deionized water (conductivity ≤ 5μS/cm) before connecting the cabinet to the cooling system. The cleaning process should not exceed 8 hours, and cleaning water should not remain in the system overnight. After cleaning, deionized water

must be completely drained before injecting coolant for circulation. Incomplete drainage of deionized water may affect the coolant's performance and parameters. Excessive impurities left in the system can cause corrosion and blockage of the cold plate or joints, resulting in product damage.

7. Improper insertion of the quick connector, which can lead to water leakage and failure of the server's over-temperature protection, causing damage to the product.

8. Lack of an exhaust device and an automatic refilling device in the pipeline system. Air mixed into the circulation system can cause the water pump to operate with gas, leading to potential damage. This can also result in overheating, burning of the server, or reduced computing power, thus damaging the product.

9. Condensation on the server: When the server is not in operation, a rise in ambient temperature from a low to a high level may lead to condensation if the server cannot adjust to the temperature change. To prevent damage, use methods such as sunlight to heat and dry the server. Failure to address condensation promptly may result in product damage.

10. Cooling water drainage: Within 2 hours of shutting down the server, when the server's inlet and outlet are disconnected from the cooling system (i.e., when the server is not circulating water), it is crucial to drain the cooling water inside the server to avoid damage to the cold plate. Neglecting this step may lead to product damage.

11. Legal Information: This warranty provides specific legal rights, and may also have additional legal rights that vary by country. Bitdeer reserves the right to interpret this warranty policy.

13. After-sales warranty Fee terms

1. If your product is deemed non-repairable, or if you opt not to repair or return a scrapped or mixed-board product, you will be responsible for the round-trip shipping costs associated with the repair order. Additionally, for products requiring repair, you will be responsible for the shipping costs to send the product to our designated address, while we will cover the shipping costs to send the repaired or replaced product back to your designated address. We do not cover any other costs, such as tariffs.

2. Please ensure that the product is sent to our designated address via a prepaid postage service. We cannot accept items sent with unpaid postage

or to non-designated addresses (including logistics pick-up points), and any consequences of these errors will be borne by you.

3. The repaired or replaced product will be shipped according to the delivery information you or your designated contact provided in the work order. If the provided delivery information is incorrect or incomplete, you will be responsible for any additional costs incurred.

4. In the event of a Dead on Arrival (DOA) situation or secondary repair where the product is non-repairable or not eligible for free repair, you may apply for a freight subsidy after sending the product to us by post. The subsidy amount will not exceed the official logistics fee standard (insurance costs are the customer's responsibility). To qualify, you must provide us with a valid receipt for the shipping costs. The freight receipt must not be forged, altered, smeared, or tampered with in any way, and it must clearly show the total freight amount, which must align with the official logistics fee standard. If the receipt does not meet these requirements, we reserve the right to deny part or all of your freight subsidy request.

5. Please ensure that products eligible for a freight subsidy are mailed separately. If you send a package that includes both products eligible for a subsidy and those that are not, we will not be able to calculate the freight for the eligible items individually, and therefore, we will not issue any subsidy for that package.

6. The risk of damage or loss during the return process transfers to you once the product is handed over to the logistics company. If any damage or loss occurs during transit, you will be responsible for resolving any disputes directly with the logistics provider.

7. Any matters not covered in this clause shall be handled in accordance with the relevant regulations of Bitdeer. Bitdeer reserves the final right of interpretation of this clause.

14. Regulatory and Compliance information

14.1 United States Federal Communications Commission (FCC)

FCC Compliance Statement

This device complies with Part 15 of the FCC Rules. Operation is subject to the

following two conditions:

(1) this device may not cause harmful interference, and

(2) this device must accept any interference received, including interference that may cause undesired operation.

Class A Digital Device Notice

Note: This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to Part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications. Operation of this equipment in a residential area is likely to cause harmful interference, in which case the user will be required to correct the interference at their own expense.

Mandatory Shielding & Grounding

To ensure continued compliance with FCC Class A emission limits, shielded cables must be used with this equipment. Use of non-shielded cables may result in interference exceeding legal limits. The equipment must be connected to a power outlet with a protective earthing (grounding) connection.

FCC Caution

Any changes or modifications not expressly approved by Bitdeer could void the user's authority to operate the equipment.

Supplier's Declaration of Conformity (SDoC)

FCC Supplier's Declaration of Conformity

Trade Name: SEALMINER

Model Number: Hydro Series

Responsible Part---U.S. Contact Information

Company: Bitdeer Sales (USA) Inc.

Street Address: 2880 Zanker Rd., Suite 205

City, State: San Jose, CA

Zip Code: CA 95134

Telephone number: +1 408-451-5200

Internet contact information: https://www.bitdeer.com/

FCC Compliance Statement:

This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions: (1) This device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation.

The FCC Supplier's Declaration of Conformity (SDoC) provided herein applies to the following models within the Hydro Series: A2 Hyd, A2 Pro Hydro, A3 Hydro, A3 Pro Hydro, A4 Pro Hydro, and A4 Ultra Hydro.

14.2 Innovation, Science and Economic Development (ISED) Canada

(English) This Class A digital apparatus complies with Canadian ICES-003 Issue 7.

(Français) Cet appareil numérique de la classe A est conforme à la norme NMB-003, version 7, du Canada.

Compliance Statement (English):

This device contains licence-exempt transmitter(s)/receiver(s) that comply with Innovation, Science and Economic Development Canada's licence-exempt RSS(s). Operation is subject to the following two conditions:

1. This device may not cause interference.
2. This device must accept any interference, including interference that may cause undesired operation of the device.

Déclaration de Conformité (Français):

L'émetteur/récepteur exempt de licence contenu dans le présent appareil est conforme aux CNR d'Innovation, Sciences et Développement économique Canada applicables aux appareils radio exempts de licence. L'exploitation est autorisée aux deux conditions suivantes :

1. L'appareil ne doit pas produire de brouillage;
2. L'appareil doit accepter tout brouillage radioélectrique subi, même si le brouillage est susceptible d'en compromettre le fonctionnement.

Labelling Identification:

CAN ICES-003 (A) / NMB-003 (A)

14.3 European Union (CE) & WEEE

SEALMINER

CE

EU Declaration of Conformity

We
Manufacturer's Name: Bitdeer Semiconductor Technology Pte. Ltd.
Manufacturer's Address: 63 Chulia Street, #15-01 OCBC Centre East Singapore 049514, Singapore

EU AR Name: Bitdeer Norway AS, c/o Javelin Corporate Services AS
EU AR Address: Vogts gate 64, 0477 Oslo, Norway
Contact: aftersales@bitdeer.com | www.bitdeer.com

declare under sole responsibility for issuing this declaration of conformity in relation to the following product:

Trademark: SEALMINER
Type of Product: Server
Model: A2 Pro Hydro
Batch Number:
Serial Number:



Product Description:
High-efficiency SHA-256 ASIC Miner for BTC/BCH/BSV coin mining.
Rating: Input: 380-480VAC, 3~ , 50/60Hz, 17A MAX

For the above given Product is hereby declared that it conforms to the essential requirements set out in community harmonization legislation mentioned below:

- Electromagnetic Compatibility(EMC) Directive (2014/30/EU)
- Low Voltage Directive (LVD) (2014/35/EU)
- Restriction of the use of certain hazardous substances (RoHS) directive (2011/65/EU) and its amendment Directive (EU) 2015/863

Reference to the relevant harmonised standards used, or references to the specifications in relation to which conformity is declared:

EMC standard:	EN 55032:2015+A11+A1 EN IEC 61000-3-11:2019 EN 61000-3-12:2011+A1 EN 55035:2017+A11
Safety standard:	EN 62368-1: 2014 + A11:2017
Chemical standard:	EN IEC 63000:2018

Signed for and on behalf of:

Singapore	23 Mar 2026	
Place of Issue	Date of Issue	Name, Function, Signature

The EU Declaration of Conformity provided in this section is representative of the A2 Pro Hydro model. For the Declaration of Conformity for other models within the Hydro Series, please contact Bitdeer Customer Support.

WEEE (Waste Electrical and Electronic Equipment)

The WEEE symbol below indicates that this product must not be disposed of with household waste. In accordance with EU Directive 2012/19/EU, this product must be handed over to an authorized collection point for the recycling of electrical and electronic equipment (EEE).

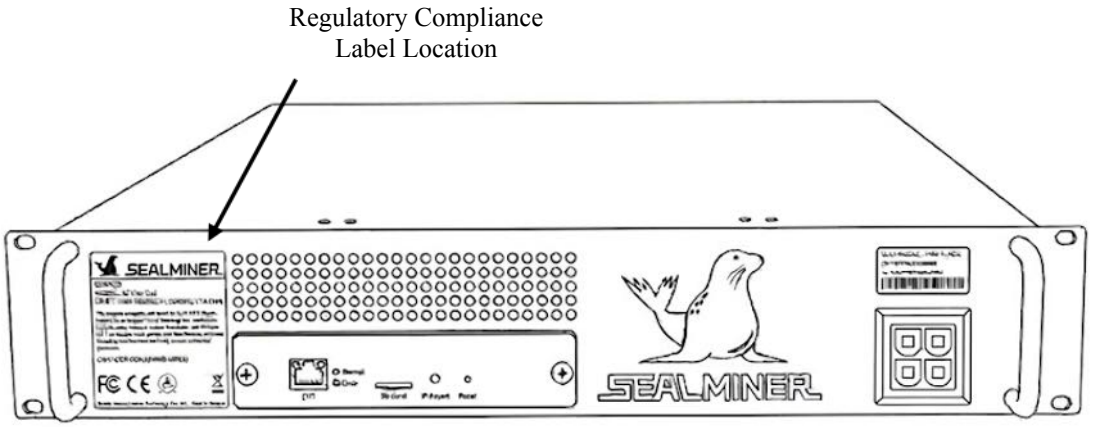
Symbol	Instructions
	The symbol shown (crossed-out wheelie bin) is on the product or its packaging, which indicates that this product must not be disposed of with other waste. Instead, it is the user's responsibility to dispose of their waste equipment by handing it over to a designated collection point for the recycling of waste electrical and electronic equipment. The separate collection and recycling of your waste equipment at the time of disposal will help to conserve natural resources and ensure that it is recycled in a manner that protects human health and the environment. For more information about where you can drop off your waste equipment for recycling, contact your local waste disposal service or Bitdeer representative for recycling information.

14.4 Safety & Power Specifications

- **Installation:** The socket-outlet must be installed near the equipment and remain easily accessible as the primary disconnect device.
- **Power Cord:** Use only the power cords provided or specified by Bitdeer. To ensure continued compliance with emission limits, shielded cables must be used.
- **Earthing:** The equipment must be connected to a power outlet with a protective earthing (grounding) connection to prevent electric shock.
- **Operating Environment:** Industrial use only. Not intended for use in locations where children are likely to be present.
- The circuit breaker on the PDU serves as the disconnect device for the server, with a rated breaking current of 20A.

14.5 Product Identification & Labelling

The regulatory label is affixed to the chassis of the device (typically on the side panel). This label contains critical information including the Model Number, Serial Number, and required certification marks (FCC, CE, IC, cTUVus & WEEE).



Sample Regulatory Label

Below is a representation of the compliance label found on the device:

Trademark: **Type of Product:** **Model:** **Input:** [Voltage/Current Specs]
FCC Compliance Statement: This device complies with part 15 of the FCC Rules. Operation is subject to the following two conditions: (1) This device may not cause harmful interference. (2) This device must accept any interference received, including interference that may cause undesired operation.
CAN ICES-003 (A) / NMB-003 (A)
[FCC Logo] [CE Logo] [cTUVus Logo] [WEEE Bin]

Manufacturer Name
Manufacturer Address

EU Authorized Representative Name
EU Authorized Representative Address

Contact information **Made in [Country]**